Notes on the Modern Theory of Linear Models

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Notation and conventions. Throughout, $\|\mathbf{v}\| = \|\mathbf{v}\|_2 = (\sum_i v_i^2)^{1/2}$ is the Euclidean norm, $\|\mathbf{A}\|_{\mathrm{op}} = \sup_{\|\mathbf{v}\|=1} \|\mathbf{A}\mathbf{v}\|$ is the operator (spectral) norm, $\|\mathbf{A}\|_{\mathrm{F}} = (\mathrm{tr}(\mathbf{A}'\mathbf{A}))^{1/2}$ is the Frobenius norm, and $\sigma_{\min}(\mathbf{A})$, $\sigma_{\max}(\mathbf{A})$ denote the smallest and largest singular values of $\mathbf{A}$. For a symmetric matrix $\mathbf{A}$, $\lambda_{\min}(\mathbf{A})$ and $\lambda_{\max}(\mathbf{A})$ denote its extreme eigenvalues; then $\|\mathbf{A}\|_{\mathrm{op}} = \max(|\lambda_{\min}(\mathbf{A})|, |\lambda_{\max}(\mathbf{A})|)$. We write $a \lesssim b$ to mean $a \leqslant Cb$ for a universal constant $C > 0$. For two symmetric matrices, $\mathbf{A} \preceq \mathbf{A}'$ means $\mathbf{A}' - \mathbf{A}$ is positive semi-definite. We write $a \wedge b = \min(a, b)$ and $a \vee b = \max(a, b)$. The standard Gaussian random vector in $\mathbb{R}^n$ is denoted $\mathbf{g} \sim N(\mathbf{0}, \mathbf{I}_n)$. In Part 3, p replaces $k+1$ for the total number of regressors (including the intercept, if present).

Prerequisites. This document assumes familiarity with the material in *Notes on the Classical Theory of Linear Models*. The primary references are R. Vershynin, *High-Dimensional Probability* (HDP, 2018), and M. J. Wainwright, *High-Dimensional Statistics* (HDS, 2019).

Part I
The Toolkit: Sub-Gaussian and Sub-Exponential Variables

1 Sub-Gaussian Random Variables

1.1 Motivation: Why Go Beyond Normality?

Remark. In the classical linear model (Assumption (A5)), exact inference rests on the normality of $\boldsymbol{\varepsilon}$. The Gauss–Markov result (Theorem 2.6 in the classical notes) needs only (A1)–(A4), but the pivotal distributions t_{n-k-1} and $F_{k,n-k-1}$ demand Gaussian errors. The modern programme asks: *what minimal distributional property replaces normality to yield sharp, non-asymptotic concentration?* The answer is the sub-Gaussian condition: the tails of the random variable decay at least as fast as those of a Gaussian. This class includes all bounded random variables, all Gaussian random variables, and many other distributions encountered in practice.

1.2 Definition and Equivalent Characterizations

Definition 1.1 (Sub-Gaussian Random Variable)**.** A real-valued random variable X with $\mathsf{E}[X] = 0$ is called **sub-Gaussian** if there exists a constant $K > 0$ such that

$$\mathsf{E}\left[e^{tX}\right] \leqslant e^{K^2t^2/2} \quad \text{for all } t \in \mathbb{R}. \tag{1}$$

The **sub-Gaussian norm** (or ψ_2-norm) of X is

$$\|X\|_{\psi_2} = \inf\left\{t > 0 : \mathsf{E}\left[e^{X^2/t^2}\right] \leqslant 2\right\}. \tag{2}$$

Theorem 1.1 (Equivalent Characterizations of Sub-Gaussianity)**.** Let X be a random variable with $\mathsf{E}[X] = 0$. The following properties are equivalent, in the sense that if one holds with constant K_i, then each of the others holds with a constant $K_j = C_{ij}K_i$ for a universal factor $C_{ij} > 0$:

(i) **MGF bound**: $\mathsf{E}[e^{tX}] \leqslant e^{K_1^2t^2/2}$ for all $t \in \mathbb{R}$.

(ii) **Tail bound**: $\mathsf{P}(|X| \geqslant t) \leqslant 2\,e^{-t^2/(2K_2^2)}$ for all $t \geqslant 0$.

(iii) **Moment bound**: $(\mathsf{E}[|X|^p])^{1/p} \leqslant K_3\sqrt{p}$ for all $p \geqslant 1$.

(iv) **ψ_2-norm finite**: $\|X\|_{\psi_2} \leqslant K_4$.

Moreover, $\|X\|_{\psi_2}$ is equivalent (up to universal constants) to each of the optimal K_i.

Proof. We prove the cycle (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) $\Rightarrow$ (i).

(i) $\Rightarrow$ (ii). By Markov's inequality and (i), for any $t > 0$ and $\lambda > 0$:

$$\mathsf{P}(X \geqslant t) = \mathsf{P}(e^{\lambda X} \geqslant e^{\lambda t}) \leqslant e^{-\lambda t}\,\mathsf{E}[e^{\lambda X}] \leqslant e^{-\lambda t + K_1^2\lambda^2/2}.$$

Optimizing over λ: set $\frac{d}{d\lambda}(-\lambda t + K_1^2\lambda^2/2) = -t + K_1^2\lambda = 0$, giving $\lambda^* = t/K_1^2$. Substituting:

$$\mathsf{P}(X \geqslant t) \leqslant e^{-t^2/(2K_1^2)}.$$

Applying the same argument to $-X$ (which also satisfies (i) with the same constant, since $\mathsf{E}[e^{-tX}] = \mathsf{E}[e^{t(-X)}]$ and $(-X)$ has mean zero):

$$\mathsf{P}(|X| \geqslant t) \leqslant 2\,e^{-t^2/(2K_1^2)}.$$

So $K_2 = K_1$ works.

(ii) $\Rightarrow$ (iii). For any $p \geqslant 1$, the layer-cake formula gives:

$$\mathsf{E}[|X|^p] = \int_0^\infty p\, t^{p-1} \mathsf{P}(|X| \geqslant t)\, \mathrm{d}t \leqslant \int_0^\infty p\, t^{p-1} \cdot 2\, e^{-t^2/(2K_2^2)}\, \mathrm{d}t. \tag{3}$$

Substituting $s = t^2/(2K_2^2)$, so $t = K_2\sqrt{2s}$ and $\mathrm{d}t = K_2\, \mathrm{d}s/\sqrt{2s}$:

$$\mathsf{E}[|X|^p] \leqslant 2p \int_0^\infty (K_2\sqrt{2s})^{p-1}\, e^{-s}\, \frac{K_2}{\sqrt{2s}}\, \mathrm{d}s = p\, (K_2\sqrt{2})^p\, \Gamma(p/2).$$

We bound $\Gamma(p/2)$. By the integral representation, $\Gamma(\alpha) = \int_0^\infty u^{\alpha-1}e^{-u}\, \mathrm{d}u$. The integrand $u^{\alpha-1}e^{-u}$ is maximized at $u = \alpha-1$, giving the crude bound $\Gamma(\alpha) \leqslant (\alpha-1)^{\alpha-1}e^{-(\alpha-1)} \int_0^\infty e^{-(u-\alpha+1)^2/(2(\alpha-1))} \sqrt{2\pi(\alpha-1)} \cdots$ Instead, we use $\Gamma(\alpha) \leqslant \alpha^{\alpha-1}$ for $\alpha \geqslant 1$ (which follows from $\Gamma(\alpha) \leqslant \alpha^\alpha/\alpha = \alpha^{\alpha-1}$ since $\Gamma(\alpha+1) = \alpha\Gamma(\alpha) \leqslant \alpha^\alpha$ by the bound $\Gamma(n+1) = n! \leqslant n^n$, extended to non-integers via log-convexity). Hence $\Gamma(p/2) \leqslant (p/2)^{p/2-1}$ for $p \geqslant 2$. Substituting:

$$\mathsf{E}[|X|^p] \leqslant p\, (K_2\sqrt{2})^p\, (p/2)^{p/2-1} \leqslant p^{p/2} \cdot 2^{p/2} \cdot K_2^p \cdot \frac{p}{(p/2)} = 2 \cdot (2K_2^2)^{p/2} \cdot p^{p/2}.$$

Taking p-th roots: $(\mathsf{E}[|X|^p])^{1/p} \leqslant 2^{1/p}\sqrt{2}\, K_2\sqrt{p} \leqslant 2\sqrt{2}\, K_2\sqrt{p}$. For $p = 1$: $\mathsf{E}[|X|] \leqslant 2\int_0^\infty e^{-t^2/(2K_2^2)}\, \mathrm{d}t = K_2\sqrt{2\pi} \leqslant 3K_2$. So $K_3 = 3K_2$ works uniformly for all $p \geqslant 1$.

(iii) $\Rightarrow$ (iv). From (iii), $\mathsf{E}[|X|^{2k}] \leqslant (K_3\sqrt{2k})^{2k} = K_3^{2k}(2k)^k$. Using the Taylor expansion:

$$\mathsf{E}\left[e^{X^2/t^2}\right] = \sum_{k=0}^\infty \frac{\mathsf{E}[|X|^{2k}]}{k!\, t^{2k}} \leqslant 1 + \sum_{k=1}^\infty \frac{K_3^{2k}(2k)^k}{k!\, t^{2k}}.$$

Since $k! \geqslant (k/e)^k$ (Stirling's lower bound), $k^k/k! \leqslant e^k$. Hence $(2k)^k/k! = 2^k \cdot k^k/k! \leqslant (2e)^k$, and:

$$\frac{K_3^{2k}(2k)^k}{k!\, t^{2k}} \leqslant \left(\frac{2eK_3^2}{t^2}\right)^k.$$

Choosing $t^2 = 4eK_3^2$: each term is $\leqslant 2^{-k}$, so $\mathsf{E}[e^{X^2/t^2}] \leqslant 1 + 1 = 2$. Hence $\|X\|_{\psi_2} \leqslant 2\sqrt{e}\, K_3$.

(iv) $\Rightarrow$ (i). Let $K = \|X\|_{\psi_2}$, so $\mathsf{E}[e^{X^2/K^2}] \leqslant 2$.

Step 1: Obtain a tail bound. By Markov's inequality: for $u \geqslant 0$,

$$\mathsf{P}(|X| \geqslant u) = \mathsf{P}(e^{X^2/K^2} \geqslant e^{u^2/K^2}) \leqslant e^{-u^2/K^2}\, \mathsf{E}[e^{X^2/K^2}] \leqslant 2e^{-u^2/K^2}.$$

This is the tail bound (ii) with $K_2 = K/\sqrt{2}$.

Step 2: Obtain moment bounds. By the implication (ii) $\Rightarrow$ (iii) (already proved), $(\mathsf{E}[|X|^p])^{1/p} \leqslant CK\sqrt{p}$ for all $p \geqslant 1$, hence $|\mathsf{E}[X^p]| \leqslant \mathsf{E}[|X|^p] \leqslant (CK)^p p^{p/2}$.

Step 3: Bound the MGF via Taylor series. Since $\mathsf{E}[X] = 0$:

$$\mathsf{E}[e^{tX}] = 1 + \sum_{p=2}^\infty \frac{t^p\, \mathsf{E}[X^p]}{p!} \leqslant 1 + \sum_{p=2}^\infty \frac{|t|^p (CK)^p\, p^{p/2}}{p!}.$$

Using $p^{p/2}/p! \leqslant p^{p/2}/(p/e)^p = e^p/p^{p/2}$ (from Stirling's lower bound $p! \geqslant (p/e)^p$):

$$\frac{(CK|t|)^p\, p^{p/2}}{p!} \leqslant (CeK|t|)^p \cdot p^{-p/2} \cdot p^{p/2} \cdot p^{-p/2} = \left(\frac{CeK|t|}{\sqrt{p}}\right)^p \cdot p^{p/2}.$$

This bound is not tight enough. Instead, use the direct estimate: $p^{p/2}/p! \leqslant e^p/p^{p/2}$, so

$$\frac{(CK|t|)^p p^{p/2}}{p!} \leqslant \frac{(CK|t|)^p e^p}{p^{p/2}} = \left(\frac{CeK|t|}{p^{1/2}}\right)^p.$$

For $p \geqslant 2$ and $|t| \leqslant 1/(2CeK)$: $CeK|t|/\sqrt{p} \leqslant CeK|t| \leqslant 1/2$, so each term is $\leqslant 2^{-p}$. Hence:

$$\mathsf{E}[e^{tX}] \leqslant 1 + \sum_{p=2}^{\infty} 2^{-p} = 1 + 1/2 < 2 \leqslant e^1, \quad \text{for } |t| \leqslant \frac{1}{2CeK}.$$

Step 4: Extend to all t. For $|t| > 1/(2CeK)$, we use the tail bound from Step 1. Split the MGF:

$$\mathsf{E}[e^{tX}] = \mathsf{E}[e^{tX}\mathbf{1}_{|X|\leqslant R}] + \mathsf{E}[e^{tX}\mathbf{1}_{|X|>R}]$$

with $R = K^2|t|$. For the first term: $e^{tX}\mathbf{1}_{|X|\leqslant R} \leqslant e^{|t|R} = e^{K^2t^2}$. For the second:

$$\begin{aligned}\mathsf{E}[e^{tX}\mathbf{1}_{|X|>R}] \leqslant \mathsf{E}[e^{|t||X|}\mathbf{1}_{|X|>R}] &= |t|\int_R^{\infty} e^{|t|u}\mathsf{P}(|X| \geqslant u)\,\mathrm{d}u + e^{|t|R}\mathsf{P}(|X| \geqslant R) \\ &\leqslant |t|\int_R^{\infty} e^{|t|u} \cdot 2e^{-u^2/K^2}\,\mathrm{d}u + 2e^{|t|R-R^2/K^2}.\end{aligned}$$

In the exponent: $|t|u - u^2/K^2 = -(u/K - K|t|/2)^2 + K^2t^2/4$. So all terms are bounded by $Ce^{K^2t^2/4}\cdot$ (polynomial in $|t|K$), giving $\mathsf{E}[e^{tX}] \leqslant e^{C'K^2t^2}$ for a universal C'.

Combining Steps 3–4: $\mathsf{E}[e^{tX}] \leqslant e^{C''K^2t^2/2}$ for all $t \in \mathbb{R}$, with $K_1 = \sqrt{2C''}K_4$. □

Example 1.1 (Sub-Gaussian Random Variables). (1) **Gaussian.** If $X \sim N(0, \sigma^2)$, then $\mathsf{E}[e^{tX}] = e^{\sigma^2t^2/2}$, so X is sub-Gaussian with $K_1 = \sigma$. Moreover, $\|X\|_{\psi_2} = C\sigma$ for a universal C.

(2) **Bounded (Rademacher).** Let X take values ± 1 with equal probability. Then $\mathsf{E}[e^{tX}] = \cosh(t) \leqslant e^{t^2/2}$, so X is sub-Gaussian with $K_1 = 1$.

(3) **Bounded (Hoeffding's lemma).** If $a \leqslant X \leqslant b$ and $\mathsf{E}[X] = 0$, then $\mathsf{E}[e^{tX}] \leqslant e^{(b-a)^2t^2/8}$, so X is sub-Gaussian with $K_1 = (b-a)/2$. The proof is given in Lemma 2.1.

(4) **Non-example (Exponential).** If $X \sim \mathrm{Exp}(1) - 1$ (centered), then $\mathsf{E}[e^{tX}] = e^{-t}/(1-t)$ for $t < 1$, which grows faster than any e^{Kt^2} as $t \to 1^-$. Hence X is *not* sub-Gaussian; it is sub-exponential (Definition 4.1).

1.3 Properties of Sub-Gaussian Random Variables

Proposition 1.1 (Closure Properties). (i) If X is sub-Gaussian and $a \in \mathbb{R}$, then aX is sub-Gaussian with $\|aX\|_{\psi_2} = |a|\,\|X\|_{\psi_2}$.

(ii) If $X_1, \ldots, X_n$ are independent sub-Gaussian random variables with $\mathsf{E}[X_i] = 0$, then $S = \sum_{i=1}^n X_i$ is sub-Gaussian with $\|S\|_{\psi_2}^2 \leqslant C\sum_{i=1}^n \|X_i\|_{\psi_2}^2$.

Proof. **(i).** Immediate from $\mathsf{E}[e^{(aX)^2/t^2}] = \mathsf{E}[e^{X^2/(t/|a|)^2}]$, which is $\leqslant 2$ when $t/|a| \geqslant \|X\|_{\psi_2}$, i.e., $t \geqslant |a|\,\|X\|_{\psi_2}$.

(ii). By independence:

$$\mathsf{E}\left[e^{tS}\right] = \prod_{i=1}^n \mathsf{E}\left[e^{tX_i}\right] \leqslant \prod_{i=1}^n e^{K_i^2t^2/2} = \exp\left(\frac{t^2}{2}\sum_{i=1}^n K_i^2\right),$$

where $K_i = C\|X_i\|_{\psi_2}$ from the MGF characterization. Hence S is sub-Gaussian with $K^2 = \sum K_i^2$, and $\|S\|_{\psi_2}^2 \leqslant C'\sum \|X_i\|_{\psi_2}^2$. □

1.4 Variance of Sub-Gaussian Variables

Proposition 1.2. If X is sub-Gaussian with $\mathsf{E}[X] = 0$, then $\mathrm{var}(X) = \mathsf{E}[X^2] \leqslant C\|X\|_{\psi_2}^2$ for a universal constant C.

Proof. From the moment bound (Theorem 1.1(iii)) with $p = 2$: $(\mathsf{E}[X^2])^{1/2} \leqslant K_3\sqrt{2}$. Since $K_3 \leqslant C\|X\|_{\psi_2}$, squaring gives $\mathsf{E}[X^2] \leqslant 2C^2\|X\|_{\psi_2}^2$. □

2 Hoeffding's Inequality

2.1 Hoeffding's Lemma

Lemma 2.1 (Hoeffding's Lemma)**.** Let X be a random variable with $\mathsf{E}[X] = 0$ and $a \leqslant X \leqslant b$ almost surely. Then for all $t \in \mathbb{R}$:

$$\mathsf{E}[e^{tX}] \leqslant \exp\left(\frac{t^2(b-a)^2}{8}\right). \tag{4}$$

Proof. **Step 1: Convexity bound.** Since e^{tx} is convex in x, for any $x \in [a, b]$:

$$e^{tx} \leqslant \frac{b-x}{b-a}\, e^{ta} + \frac{x-a}{b-a}\, e^{tb}.$$

Taking expectations and using $\mathsf{E}[X] = 0$:

$$\mathsf{E}[e^{tX}] \leqslant \frac{b}{b-a}\, e^{ta} + \frac{-a}{b-a}\, e^{tb} = \frac{b\, e^{ta} - a\, e^{tb}}{b-a}. \tag{5}$$

Step 2: Logarithmic bound. Let $h = t(b-a)$, $\theta = -a/(b-a) \in [0, 1]$. Then $1-\theta = b/(b-a)$. Rewrite (5):

$$\mathsf{E}[e^{tX}] \leqslant (1-\theta)\, e^{-\theta h} + \theta\, e^{(1-\theta)h} = e^{-\theta h}\big[(1-\theta) + \theta\, e^{h}\big].$$

Define $\varphi(h) = -\theta h + \ln\big[(1-\theta) + \theta\, e^{h}\big]$. Then $\mathsf{E}[e^{tX}] \leqslant e^{\varphi(h)}$.

Step 3: Taylor expansion of φ. We have $\varphi(0) = 0$ and $\varphi'(0) = -\theta + \theta = 0$. Computing the second derivative:

$$\varphi''(h) = \frac{\theta(1-\theta)\, e^{h}}{[(1-\theta) + \theta\, e^{h}]^2}.$$

Let $u = \theta\, e^{h}/[(1-\theta) + \theta\, e^{h}] \in (0, 1)$. Then $\varphi''(h) = u(1-u) \leqslant 1/4$ (since $u(1-u) \leqslant 1/4$ for $u \in [0, 1]$). By Taylor's theorem with remainder:

$$\varphi(h) = \varphi(0) + h\varphi'(0) + \frac{h^2}{2}\varphi''(\xi) \leqslant \frac{h^2}{8}$$

for some ξ between 0 and h. Hence $\mathsf{E}[e^{tX}] \leqslant e^{h^2/8} = e^{t^2(b-a)^2/8}$. □

2.2 Hoeffding's Inequality

Theorem 2.1 (Hoeffding's Inequality)**.** Let $X_1, \ldots, X_n$ be independent random variables with $\mathsf{E}[X_i] = 0$ and $a_i \leqslant X_i \leqslant b_i$ almost surely for each i. Then for all $t \geqslant 0$:

$$\boxed{\mathsf{P}\left(\left|\sum_{i=1}^{n} X_i\right| \geqslant t\right) \leqslant 2\exp\left(-\frac{2t^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right).} \tag{6}$$

Proof. **Step 1: One-sided bound via Chernoff.** For $\lambda > 0$:

$$\begin{aligned}\mathsf{P}\left(\sum_{i=1}^{n} X_i \geqslant t\right) &= \mathsf{P}\big(e^{\lambda \sum X_i} \geqslant e^{\lambda t}\big) \leqslant e^{-\lambda t}\, \mathsf{E}\big[e^{\lambda \sum X_i}\big] \\ &= e^{-\lambda t} \prod_{i=1}^{n} \mathsf{E}[e^{\lambda X_i}] \leqslant e^{-\lambda t} \prod_{i=1}^{n} e^{\lambda^2 (b_i - a_i)^2/8},\end{aligned} \tag{7}$$

where the last inequality uses Hoeffding's Lemma (Lemma 2.1) applied to each X_i.

Step 2: Optimize λ. From (7):

$$\mathsf{P}\Big(\sum X_i \geqslant t\Big) \leqslant \exp\left(-\lambda t + \frac{\lambda^2}{8}\sum_{i=1}^{n}(b_i - a_i)^2\right).$$

Let $\sigma_H^2 = \frac{1}{4}\sum(b_i - a_i)^2$. Setting $\frac{d}{d\lambda}(-\lambda t + \lambda^2\sigma_H^2/2) = -t + \lambda\sigma_H^2 = 0$ gives $\lambda^* = t/\sigma_H^2$:

$$\mathsf{P}\Big(\sum X_i \geqslant t\Big) \leqslant \exp\left(-\frac{t^2}{2\sigma_H^2}\right) = \exp\left(-\frac{2t^2}{\sum(b_i - a_i)^2}\right).$$

Step 3: Two-sided bound. Apply the same argument to $-X_1, \ldots, -X_n$ to get $\mathsf{P}(\sum X_i \leqslant -t) \leqslant \exp(-2t^2/\sum(b_i - a_i)^2)$. By the union bound, $\mathsf{P}(|\sum X_i| \geqslant t) \leqslant 2\exp(-2t^2/\sum(b_i - a_i)^2)$. □

Corollary 2.1 (Hoeffding for i.i.d. Bounded Variables)**.** Let $X_1, \ldots, X_n$ be i.i.d. with $\mathsf{E}[X_i] = \mu$ and $a \leqslant X_i \leqslant b$. Then for $\bar{X} = \frac{1}{n}\sum X_i$:

$$\mathsf{P}\big(|\bar{X} - \mu| \geqslant t\big) \leqslant 2\exp\left(-\frac{2nt^2}{(b-a)^2}\right). \tag{8}$$

Proof. Apply Theorem 2.1 to $Y_i = X_i - \mu$ (so $\mathsf{E}[Y_i] = 0$, $a - \mu \leqslant Y_i \leqslant b - \mu$, and $b_i - a_i = b - a$ for all i). Then $\sum Y_i = n(\bar{X} - \mu)$, and the bound gives $\mathsf{P}(|n(\bar{X} - \mu)| \geqslant nt) \leqslant 2\exp(-2n^2t^2/(n(b-a)^2))$. □

3 Sub-Gaussian Concentration for Sums and Quadratic Forms

3.1 General Sub-Gaussian Hoeffding Inequality

Theorem 3.1 (General Hoeffding Inequality for Sub-Gaussians)**.** Let $X_1, \ldots, X_n$ be independent, mean-zero, sub-Gaussian random variables. Then for all $t \geqslant 0$:

$$\mathsf{P}\left(\left|\sum_{i=1}^{n} X_i\right| \geqslant t\right) \leqslant 2\exp\left(-\frac{ct^2}{\sum_{i=1}^{n}\|X_i\|_{\psi_2}^2}\right), \tag{9}$$

where $c > 0$ is a universal constant.

Proof. By the MGF characterization (Theorem 1.1(i)), each X_i satisfies $\mathsf{E}[e^{\lambda X_i}] \leqslant e^{C^2\|X_i\|_{\psi_2}^2\lambda^2/2}$. The proof then follows the Chernoff–optimize pattern of Theorem 2.1, with $(b_i - a_i)^2/4$ replaced by $C^2\|X_i\|_{\psi_2}^2$:

$$\mathsf{P}\Big(\sum X_i \geqslant t\Big) \leqslant \exp\left(-\lambda t + \frac{C^2\lambda^2}{2}\sum\|X_i\|_{\psi_2}^2\right).$$

Setting $\lambda^* = t/(C^2\sum\|X_i\|_{\psi_2}^2)$ gives $\mathsf{P}(\sum X_i \geqslant t) \leqslant \exp(-t^2/(2C^2\sum\|X_i\|_{\psi_2}^2))$, and the two-sided bound follows by the union bound with $c = 1/(2C^2)$. □

3.2 Sub-Gaussian Random Vectors

Definition 3.1 (Sub-Gaussian Random Vector)**.** A random vector $\mathbf{x} \in \mathbb{R}^p$ is **sub-Gaussian** if all one-dimensional marginals are sub-Gaussian. Formally, there exists $K > 0$ such that

$$\|\langle \mathbf{x}, \mathbf{v}\rangle\|_{\psi_2} \leqslant K\,\|\mathbf{v}\| \quad \text{for all } \mathbf{v} \in \mathbb{R}^p. \tag{10}$$

The smallest such K is the **sub-Gaussian norm** of $\mathbf{x}$, denoted $\|\mathbf{x}\|_{\psi_2}$.

Example 3.1. (1) If $\mathbf{x} \sim N(\mathbf{0}, \mathbf{\Sigma})$, then $\langle \mathbf{x}, \mathbf{v}\rangle \sim N(0, \mathbf{v}'\mathbf{\Sigma}\mathbf{v})$ has $\|\langle \mathbf{x}, \mathbf{v}\rangle\|_{\psi_2} = C(\mathbf{v}'\mathbf{\Sigma}\mathbf{v})^{1/2} \leqslant C\|\mathbf{\Sigma}\|_{\mathrm{op}}^{1/2}\|\mathbf{v}\|$. So $\mathbf{x}$ is sub-Gaussian with $\|\mathbf{x}\|_{\psi_2} \leqslant C\|\mathbf{\Sigma}\|_{\mathrm{op}}^{1/2}$.

(2) If $\mathbf{x} = (\varepsilon_1, \ldots, \varepsilon_p)'$ where ε_i are independent Rademacher, then $\langle \mathbf{x}, \mathbf{v}\rangle = \sum v_i\varepsilon_i$ has $\|\langle \mathbf{x}, \mathbf{v}\rangle\|_{\psi_2} \leqslant C\|\mathbf{v}\|$ by Proposition 1.1(ii).

3.3 Hanson–Wright Inequality for Quadratic Forms

Theorem 3.2 (Hanson–Wright Inequality)**.** Let $\mathbf{x} = (X_1, \ldots, X_n)' \in \mathbb{R}^n$ be a random vector with independent, mean-zero, sub-Gaussian components satisfying $\|X_i\|_{\psi_2} \leqslant K$. Let $\mathbf{A}$ be an $n \times n$ matrix. Then for all $t \geqslant 0$:

$$\boxed{\mathsf{P}(|\mathbf{x}'\mathbf{A}\mathbf{x} - \mathsf{E}[\mathbf{x}'\mathbf{A}\mathbf{x}]| \geqslant t) \leqslant 2\exp\left[-c\min\left(\frac{t^2}{K^4\|\mathbf{A}\|_{\mathrm{F}}^2}, \frac{t}{K^2\|\mathbf{A}\|_{\mathrm{op}}}\right)\right],} \tag{11}$$

where $c > 0$ is a universal constant.

Proof. **Step 1: Reduce to the zero-diagonal case.** Write $\mathbf{A} = \mathbf{D} + \mathbf{A}_0$, where $\mathbf{D} = \mathrm{diag}(a_{11}, \ldots, a_{nn})$ and $\mathbf{A}_0$ has zero diagonal. Then:

$$\mathbf{x}'\mathbf{A}\mathbf{x} - \mathsf{E}[\mathbf{x}'\mathbf{A}\mathbf{x}] = \underbrace{\sum_{i=1}^{n} a_{ii}(X_i^2 - \mathsf{E}[X_i^2])}_{\text{diagonal part}} + \underbrace{\mathbf{x}'\mathbf{A}_0\mathbf{x}}_{\text{off-diagonal part}},$$

where $\mathsf{E}[\mathbf{x}'\mathbf{A}_0\mathbf{x}] = \sum_{i \neq j} a_{ij}\mathsf{E}[X_i]\mathsf{E}[X_j] = 0$ by independence and mean zero.

Diagonal part. The variables $Y_i = X_i^2 - \mathsf{E}[X_i^2]$ are independent, mean-zero, and sub-exponential with $\|Y_i\|_{\psi_1} \leqslant CK^2$ (Proposition 4.1). Since $\sum a_{ii}Y_i$ is a weighted sum with $\|\sum a_{ii}Y_i\|_{\psi_1} \leqslant CK^2 \max|a_{ii}|$ and $\sum(a_{ii}\|Y_i\|_{\psi_1})^2 \leqslant C^2K^4 \sum a_{ii}^2 \leqslant C^2K^4\|\mathbf{A}\|_{\mathrm{F}}^2$, Bernstein's inequality (Theorem 4.1) gives:

$$\mathsf{P}\left(\left|\sum a_{ii}Y_i\right| \geqslant t\right) \leqslant 2\exp\left[-c_0\min\left(\frac{t^2}{K^4\|\mathbf{D}\|_{\mathrm{F}}^2}, \frac{t}{K^2\|\mathbf{D}\|_{\mathrm{op}}}\right)\right]. \tag{12}$$

Since $\|\mathbf{D}\|_{\mathrm{F}} \leqslant \|\mathbf{A}\|_{\mathrm{F}}$ and $\|\mathbf{D}\|_{\mathrm{op}} \leqslant \|\mathbf{A}\|_{\mathrm{op}}$, this is bounded by the right-hand side of (11) (up to constants).

Step 2: Condition on an independent copy (off-diagonal part). Let $\mathbf{y} = (Y_1, \ldots, Y_n)'$ be an independent copy of $\mathbf{x}$. We claim that for every convex increasing $F : [0, \infty) \to \mathbb{R}$:

$$\mathsf{E}[F(|\mathbf{x}'\mathbf{A}_0\mathbf{x}|)] \leqslant \mathsf{E}[F(4|\mathbf{x}'\mathbf{A}_0\mathbf{y}|)]. \tag{13}$$

This is a standard decoupling inequality. To prove it: let $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_n)$ be i.i.d. Rademacher, independent of $\mathbf{x}, \mathbf{y}$. Define $\mathbf{z}^{(\boldsymbol{\delta})} = (\delta_1 Z_1, \ldots, \delta_n Z_n)'$ where $Z_i = X_i$ if $\delta_i = 1$ and $Z_i = Y_i$ if $\delta_i = -1$. Since X_i and Y_i are i.i.d., $\mathbf{z}^{(\boldsymbol{\delta})}$ has the same distribution as $\mathbf{x}$ for every fixed $\boldsymbol{\delta}$. Hence $\mathsf{E}_{\mathbf{x}}[F(|\mathbf{x}'\mathbf{A}_0\mathbf{x}|)] = \mathsf{E}_{\mathbf{x},\mathbf{y},\boldsymbol{\delta}}[F(|\mathbf{z}^{(\boldsymbol{\delta})\prime}\mathbf{A}_0\mathbf{z}^{(\boldsymbol{\delta})}|)]$.

Now condition on $\boldsymbol{\delta}$ and expand: $\mathbf{z}^{(\boldsymbol{\delta})\prime}\mathbf{A}_0\mathbf{z}^{(\boldsymbol{\delta})} = \sum_{i \neq j}(A_0)_{ij}Z_i^{(\delta_i)}Z_j^{(\delta_j)}$. When $\delta_i = \delta_j$, $Z_i^{(\delta_i)}Z_j^{(\delta_j)}$ is either X_iX_j or Y_iY_j; when $\delta_i \neq \delta_j$, it is a cross term X_iY_j or Y_iX_j. By conditioning iteratively on $\delta_1, \ldots, \delta_n$ and using the triangle inequality at each step (since $\mathsf{E}_{\delta_k}[|g(\delta_k)|] \geqslant |g(+1) + g(-1)|/2$ for the "matching" terms, and Jensen's inequality for the convex function F), one obtains (13) with the factor $4 = 2^2$ arising from n successive conditionings each contributing at most a factor of 2, but the 2's multiply pairwise (see de la Peña and Giné, *Decoupling*, Theorem 3.1.1, for the precise combinatorial argument).

Step 3: Condition on y; apply sub-Gaussian concentration. Fix $\mathbf{y}$. The bilinear form $\mathbf{x}'\mathbf{A}_0\mathbf{y} = \sum_{i=1}^n X_i\,(\mathbf{A}_0\mathbf{y})_i$ is a sum of independent, mean-zero, sub-Gaussian random variables (in the randomness of $\mathbf{x}$) with:

$$\|X_i(\mathbf{A}_0\mathbf{y})_i\|_{\psi_2} \leqslant K|(\mathbf{A}_0\mathbf{y})_i|.$$

By Theorem 3.1:

$$\mathsf{P}\big(|\mathbf{x}'\mathbf{A}_0\mathbf{y}| \geqslant s \mid \mathbf{y}\big) \leqslant 2\exp\left(-\frac{cs^2}{K^2\|\mathbf{A}_0\mathbf{y}\|^2}\right). \tag{14}$$

Equivalently, the conditional MGF satisfies:

$$\mathsf{E}[e^{\lambda\mathbf{x}'\mathbf{A}_0\mathbf{y}} \mid \mathbf{y}] \leqslant e^{C_1K^2\lambda^2\|\mathbf{A}_0\mathbf{y}\|^2/2} \quad \text{for all } \lambda \in \mathbb{R}. \tag{15}$$

Step 4: Control $\|\mathbf{A}_0\mathbf{y}\|^2$ and integrate over $\mathbf{y}$. Taking expectation of (15) over $\mathbf{y}$:

$$\mathsf{E}[e^{\lambda\mathbf{x}'\mathbf{A}_0\mathbf{y}}] = \mathsf{E}_{\mathbf{y}}\left[\mathsf{E}_{\mathbf{x}}[e^{\lambda\mathbf{x}'\mathbf{A}_0\mathbf{y}} \mid \mathbf{y}]\right] \leqslant \mathsf{E}_{\mathbf{y}}\left[e^{C_1K^2\lambda^2\|\mathbf{A}_0\mathbf{y}\|^2/2}\right]. \tag{16}$$

Now $\|\mathbf{A}_0\mathbf{y}\|^2 = \mathbf{y}'\mathbf{A}_0'\mathbf{A}_0\mathbf{y}$ is a quadratic form in the sub-Gaussian vector $\mathbf{y}$. Its expectation is $\mathsf{E}[\|\mathbf{A}_0\mathbf{y}\|^2] = \sigma_Y^2 \operatorname{tr}(\mathbf{A}_0'\mathbf{A}_0) = \sigma_Y^2\|\mathbf{A}_0\|_{\mathrm{F}}^2$ where $\sigma_Y^2 = \mathsf{E}[Y_i^2] \leqslant CK^2$.

To bound $\mathsf{E}[e^{\mu\|\mathbf{A}_0\mathbf{y}\|^2}]$ for $\mu = C_1K^2\lambda^2/2$: write $\|\mathbf{A}_0\mathbf{y}\|^2 = \sum_{j=1}^n(\sum_i(A_0)_{ji}Y_i)^2 = \sum_j W_j^2$ where $W_j = \sum_i(A_0)_{ji}Y_i$ are sub-Gaussian (being linear combinations of independent sub-Gaussians). We need the bound only for μ small enough. Since $\|W_j\|_{\psi_2} \leqslant CK\|(\mathbf{A}_0)_{j\cdot}\|$ (the j-th row norm), and W_j^2 is sub-exponential with $\|W_j^2\|_{\psi_1} \leqslant C^2K^2\|(\mathbf{A}_0)_{j\cdot}\|^2$, the variables $W_j^2 - \mathsf{E}[W_j^2]$ are mean-zero sub-exponential with norm $\leqslant C^2K^2\|(\mathbf{A}_0)_{j\cdot}\|^2$. However, $W_1, \ldots, W_n$ are *not* independent in general (they share the Y_i's).

A cleaner route: since we need the Chernoff bound for the bilinear form and the quantity $\|\mathbf{A}_0\mathbf{y}\|^2$ only appears in the exponent, we can bound (16) directly. The quadratic form $\|\mathbf{A}_0\mathbf{y}\|^2 = \mathbf{y}'\mathbf{A}_0'\mathbf{A}_0\mathbf{y}$ is governed by the matrix $\mathbf{A}_0'\mathbf{A}_0$, whose operator norm is $\|\mathbf{A}_0'\mathbf{A}_0\|_{\mathrm{op}} = \|\mathbf{A}_0\|_{\mathrm{op}}^2$. For $\mu = C_1K^2\lambda^2/2 \leqslant c_2/(K^2\|\mathbf{A}_0\|_{\mathrm{op}}^2)$, i.e., $|\lambda| \leqslant c_3/(K^2\|\mathbf{A}_0\|_{\mathrm{op}})$:

$$\mathsf{E}[e^{\mu\|\mathbf{A}_0\mathbf{y}\|^2}] \leqslant e^{2\mu CK^2\|\mathbf{A}_0\|_{\mathrm{F}}^2},$$

which follows because for μ small, $\mathsf{E}[e^{\mu Z}] \leqslant e^{2\mu\mathsf{E}[Z]}$ when $Z = \|\mathbf{A}_0\mathbf{y}\|^2$ is a non-negative random variable concentrating around its mean (by a Bernstein-type argument on the Taylor series $\mathsf{E}[e^{\mu Z}] = 1 + \mu\mathsf{E}[Z] + \sum_{k\geqslant 2}\mu^k\mathsf{E}[Z^k]/k!$, using $\mathsf{E}[Z^k] \leqslant k!\,(CK^2)^{2k-2}\mathsf{E}[Z^2]\cdot\|\mathbf{A}_0\|_{\mathrm{op}}^{2(k-1)}$ and geometric series convergence for $\mu \leqslant c/(K^2\|\mathbf{A}_0\|_{\mathrm{op}}^2)$).

Substituting back into (16):

$$\mathsf{E}[e^{\lambda\mathbf{x}'\mathbf{A}_0\mathbf{y}}] \leqslant e^{C_2K^4\lambda^2\|\mathbf{A}_0\|_{\mathrm{F}}^2} \quad \text{for } |\lambda| \leqslant \frac{c_3}{K^2\|\mathbf{A}_0\|_{\mathrm{op}}}.$$

Step 5: Chernoff bound with constrained optimization. For $t > 0$:

$$\mathsf{P}(\mathbf{x}'\mathbf{A}_0\mathbf{y} \geqslant t) \leqslant \inf_{0<\lambda\leqslant c_3/(K^2\|\mathbf{A}_0\|_{\mathrm{op}})} \exp(-\lambda t + C_2K^4\lambda^2\|\mathbf{A}_0\|_{\mathrm{F}}^2).$$

The unconstrained optimum is $\lambda^* = t/(2C_2K^4\|\mathbf{A}_0\|_{\mathrm{F}}^2)$.

Sub-Gaussian regime (λ^* satisfies the constraint, i.e., $t \leqslant 2C_2c_3K^2\|\mathbf{A}_0\|_{\mathrm{F}}^2/\|\mathbf{A}_0\|_{\mathrm{op}}$):

$$\mathsf{P}(\mathbf{x}'\mathbf{A}_0\mathbf{y} \geqslant t) \leqslant \exp\left(-\frac{t^2}{4C_2K^4\|\mathbf{A}_0\|_{\mathrm{F}}^2}\right).$$

Sub-exponential regime (λ^* violates the constraint): set $\lambda = c_3/(K^2\|\mathbf{A}_0\|_{\mathrm{op}})$. Since $t > 2C_2c_3K^2\|\mathbf{A}_0\|_{\mathrm{F}}^2/\|\mathbf{A}_0$ the linear term $-\lambda t$ dominates:

$$\mathsf{P}(\mathbf{x}'\mathbf{A}_0\mathbf{y} \geqslant t) \leqslant \exp\left(-\frac{c_3 t}{2K^2\|\mathbf{A}_0\|_{\mathrm{op}}}\right).$$

Step 6: From $\mathbf{x}'\mathbf{A}_0\mathbf{y}$ back to $\mathbf{x}'\mathbf{A}_0\mathbf{x}$. By the decoupling inequality (13) with $F(u) = \mathbf{1}\{u \geqslant t\}$ (which is increasing, and the inequality extends to indicators via monotone approximation):

$$\mathsf{P}(|\mathbf{x}'\mathbf{A}_0\mathbf{x}| \geqslant t) \leqslant \mathsf{P}(4|\mathbf{x}'\mathbf{A}_0\mathbf{y}| \geqslant t) = \mathsf{P}(|\mathbf{x}'\mathbf{A}_0\mathbf{y}| \geqslant t/4).$$

Replacing t by $t/4$ in Steps 5, and using $\|\mathbf{A}_0\|_{\mathrm{F}} \leqslant \|\mathbf{A}\|_{\mathrm{F}}$, $\|\mathbf{A}_0\|_{\mathrm{op}} \leqslant 2\|\mathbf{A}\|_{\mathrm{op}}$ (since $\|\mathbf{D}\|_{\mathrm{op}} \leqslant \|\mathbf{A}\|_{\mathrm{op}}$):

$$\mathsf{P}(|\mathbf{x}'\mathbf{A}_0\mathbf{x}| \geqslant t) \leqslant 2\exp\left[-c\min\left(\frac{t^2}{K^4\|\mathbf{A}\|_{\mathrm{F}}^2}, \frac{t}{K^2\|\mathbf{A}\|_{\mathrm{op}}}\right)\right].$$

Step 7: Combine diagonal and off-diagonal. By the triangle inequality:

$$|\mathbf{x}'\mathbf{A}\mathbf{x} - \mathsf{E}[\mathbf{x}'\mathbf{A}\mathbf{x}]| \leqslant \left|\sum a_{ii}Y_i\right| + |\mathbf{x}'\mathbf{A}_0\mathbf{x}|.$$

Using $\mathsf{P}(|U+V| \geqslant t) \leqslant \mathsf{P}(|U| \geqslant t/2) + \mathsf{P}(|V| \geqslant t/2)$ and substituting (12) and the bound from Step 6 with $t/2$:

$$\mathsf{P}(|\mathbf{x}'\mathbf{A}\mathbf{x} - \mathsf{E}[\mathbf{x}'\mathbf{A}\mathbf{x}]| \geqslant t) \leqslant 2\exp\left[-c' \min\left(\frac{t^2}{K^4\|\mathbf{A}\|_{\mathrm{F}}^2}, \frac{t}{K^2\|\mathbf{A}\|_{\mathrm{op}}}\right)\right]$$

with a possibly smaller constant $c' > 0$. □

Remark. When $\mathbf{A}$ is symmetric idempotent (as in the classical theory: $\mathbf{A} = \mathbf{H}$ or $\mathbf{A} = \mathbf{M}$), $\|\mathbf{A}\|_{\mathrm{F}}^2 = \mathrm{tr}(\mathbf{A}'\mathbf{A}) = \mathrm{tr}(\mathbf{A}) = \mathrm{rank}(\mathbf{A}) = r$, and $\|\mathbf{A}\|_{\mathrm{op}} = 1$. The Hanson–Wright inequality then gives

$$\mathsf{P}(|\mathbf{x}'\mathbf{A}\mathbf{x} - \mathsf{E}[\mathbf{x}'\mathbf{A}\mathbf{x}]| \geqslant t) \leqslant 2\exp\left[-c \min\left(\frac{t^2}{K^4 r}, \frac{t}{K^2}\right)\right].$$

Compare this with the exact chi-square distribution $\mathbf{z}'\mathbf{A}\mathbf{z} \sim \chi_r^2$ under Gaussianity (Theorem 4.3 in the classical notes). The Hanson–Wright bound gives the correct "scale" r (in the Gaussian regime, χ_r^2 has variance $2r$) and extends to sub-Gaussian errors with no distributional assumption.

4 Sub-Exponential Variables and Bernstein's Inequality

4.1 Definition and Basic Properties

Definition 4.1 (Sub-Exponential Random Variable)**.** A random variable X is called **sub-exponential** if there exist constants $K_1, K_2 > 0$ such that

$$\mathsf{E}[e^{tX}] \leqslant e^{K_1^2 t^2/2} \quad \text{for all } |t| \leqslant 1/K_2. \tag{17}$$

The **sub-exponential norm** (or ψ_1-norm) is

$$\|X\|_{\psi_1} = \inf\left\{t > 0 : \mathsf{E}\left[e^{|X|/t}\right] \leqslant 2\right\}. \tag{18}$$

Proposition 4.1 (Squares of Sub-Gaussians Are Sub-Exponential)**.** If X is sub-Gaussian, then X^2 is sub-exponential with $\|X^2\|_{\psi_1} \leqslant C\|X\|_{\psi_2}^2$.

Proof. We have $\mathsf{E}[e^{X^2/t}] = \mathsf{E}[e^{|X|^2/t}]$. By the definition of $\|X\|_{\psi_2}$, $\mathsf{E}[e^{X^2/\|X\|_{\psi_2}^2}] \leqslant 2$. Setting $t = \|X\|_{\psi_2}^2$, we get $\mathsf{E}[e^{|X^2|/t}] \leqslant 2$. Hence $\|X^2\|_{\psi_1} \leqslant \|X\|_{\psi_2}^2$. □

Example 4.1. (1) χ_1^2: if $Z \sim N(0,1)$, then Z^2 is sub-exponential by Proposition 4.1. More generally, $\chi_n^2 = \sum_{i=1}^n Z_i^2$ is sub-exponential.

(2) $\mathrm{Exp}(\lambda)$: if $X \sim \mathrm{Exp}(\lambda)$, then $\|X\|_{\psi_1} = 1/\lambda$.

(3) Products: if X, Y are sub-Gaussian, then XY is sub-exponential with $\|XY\|_{\psi_1} \leqslant \|X\|_{\psi_2}\|Y\|_{\psi_2}$.

4.2 Bernstein's Inequality

Theorem 4.1 (Bernstein's Inequality)**.** Let $X_1, \ldots, X_n$ be independent, mean-zero, sub-exponential random variables. Then for all $t \geqslant 0$:

$$\boxed{\mathsf{P}\left(\left|\sum_{i=1}^n X_i\right| \geqslant t\right) \leqslant 2\exp\left[-c \min\left(\frac{t^2}{\sum_{i=1}^n \|X_i\|_{\psi_1}^2}, \frac{t}{\max_i \|X_i\|_{\psi_1}}\right)\right],} \tag{19}$$

where $c > 0$ is a universal constant.

Proof. For $|\lambda| \leqslant 1/(C \max_i \|X_i\|_{\psi_1})$, each factor satisfies $\mathsf{E}[e^{\lambda X_i}] \leqslant e^{C'^2 \|X_i\|_{\psi_1}^2 \lambda^2/2}$ by the sub-exponential MGF condition. The Chernoff argument gives:

$$\mathsf{P}\Big(\sum X_i \geqslant t\Big) \leqslant \exp\Big(-\lambda t + \frac{C'^2 \lambda^2}{2} \sum \|X_i\|_{\psi_1}^2\Big).$$

We optimize subject to the constraint $\lambda \leqslant 1/(C \max_i \|X_i\|_{\psi_1})$. The unconstrained optimum is $\lambda^* = t/(C'^2 \sum \|X_i\|_{\psi_1}^2)$.

Case 1: λ^* satisfies the constraint, i.e., $t \leqslant C'^2 \sum \|X_i\|_{\psi_1}^2/(C \max_i \|X_i\|_{\psi_1})$. Substituting λ^* gives the sub-Gaussian regime: $\exp(-t^2/(2C'^2 \sum \|X_i\|_{\psi_1}^2))$.

Case 2: λ^* violates the constraint. Set $\lambda = 1/(C \max_i \|X_i\|_{\psi_1})$:

$$\mathsf{P}\Big(\sum X_i \geqslant t\Big) \leqslant \exp\Big(-\frac{t}{C \max_i \|X_i\|_{\psi_1}} + \frac{C'^2 \sum \|X_i\|_{\psi_1}^2}{2C^2 (\max_i \|X_i\|_{\psi_1})^2}\Big).$$

Since $\sum \|X_i\|_{\psi_1}^2 \leqslant n(\max_i \|X_i\|_{\psi_1})^2$ and the second term is $O(n)$ while the first grows linearly in t, for t in this range the bound is dominated by the linear term: $\exp(-ct/\max_i \|X_i\|_{\psi_1})$.

The minimum in (19) captures both cases. The two-sided bound follows by the union bound. □

Remark (Comparison: Hoeffding vs. Bernstein)**.** For bounded variables $|X_i| \leqslant M$ (hence sub-Gaussian and sub-exponential), both inequalities apply. Hoeffding gives purely Gaussian tails $\sim e^{-ct^2}$. Bernstein also gives Gaussian tails for small t, but transitions to exponential tails $\sim e^{-ct}$ for $t \gtrsim \sum \|X_i\|_{\psi_1}^2/\max_i \|X_i\|_{\psi_1}$. Bernstein can be sharper because its Gaussian regime uses the variance $\sum \|X_i\|_{\psi_1}^2$, which can be much smaller than the range-based quantity $\sum (b_i - a_i)^2$ in Hoeffding's bound when the variables are concentrated near zero.

Part II
Random Matrices: Covariance Estimation and Design Conditioning

5 Covering Numbers and ε-Nets

Definition 5.1 (Covering Number and ε-Net)**.** Let (T, d) be a metric space and $\varepsilon > 0$. An **ε-net** of T is a subset $\mathcal{N} \subseteq T$ such that for every $\mathbf{v} \in T$ there exists $\mathbf{u} \in \mathcal{N}$ with $d(\mathbf{v}, \mathbf{u}) \leqslant \varepsilon$. The **covering number** $N(T, d, \varepsilon)$ is the smallest cardinality of an ε-net of T.

Lemma 5.1 (Covering Number of the Unit Sphere)**.** For every $\varepsilon \in (0, 1)$:

$$N(S^{n-1}, \|\cdot\|, \varepsilon) \leqslant \left(\frac{2}{\varepsilon} + 1\right)^n. \tag{20}$$

In particular, $N(S^{n-1}, \|\cdot\|, 1/4) \leqslant 9^n$.

Proof. Let $\mathcal{N}$ be a maximal ε-separated subset of S^{n-1}: $\|\mathbf{u} - \mathbf{v}\| > \varepsilon$ for all distinct $\mathbf{u}, \mathbf{v} \in \mathcal{N}$, and $\mathcal{N}$ cannot be enlarged. Then $\mathcal{N}$ is an ε-net (otherwise some point has distance $> \varepsilon$ from every element of $\mathcal{N}$, contradicting maximality). The open balls $B(\mathbf{u}, \varepsilon/2)$ for $\mathbf{u} \in \mathcal{N}$ are pairwise disjoint (if $\mathbf{w} \in B(\mathbf{u}_1, \varepsilon/2) \cap B(\mathbf{u}_2, \varepsilon/2)$, then $\|\mathbf{u}_1 - \mathbf{u}_2\| \leqslant \|\mathbf{u}_1 - \mathbf{w}\| + \|\mathbf{w} - \mathbf{u}_2\| < \varepsilon$, contradicting separation). All such balls lie in $B(\mathbf{0}, 1 + \varepsilon/2)$. Comparing volumes ($\mathrm{vol}(B(\mathbf{0}, r)) = c_n r^n$):

$$|\mathcal{N}| \cdot (\varepsilon/2)^n \leqslant (1 + \varepsilon/2)^n \implies |\mathcal{N}| \leqslant \left(\frac{2 + \varepsilon}{\varepsilon}\right)^n = \left(\frac{2}{\varepsilon} + 1\right)^n. \qquad \square$$

Lemma 5.2 (ε-Net Approximation for Operator Norms)**.** Let $\mathbf{A}$ be a symmetric $n \times n$ matrix and $\mathcal{N}$ a $(1/4)$-net of S^{n-1}. Then:

$$\|\mathbf{A}\|_{\mathrm{op}} \leqslant 2 \sup_{\mathbf{v} \in \mathcal{N}} |\mathbf{v}'\mathbf{A}\mathbf{v}|. \tag{21}$$

Proof. Since $\mathbf{A}$ is symmetric, $\|\mathbf{A}\|_{\mathrm{op}} = \sup_{\mathbf{v} \in S^{n-1}} |\mathbf{v}'\mathbf{A}\mathbf{v}|$. Let $\mathbf{v}^*$ achieve this supremum and choose $\mathbf{v}_0 \in \mathcal{N}$ with $\|\mathbf{v}^* - \mathbf{v}_0\| \leqslant 1/4$. Then:

$$\begin{aligned} |\mathbf{v}^{*\prime}\mathbf{A}\mathbf{v}^* - \mathbf{v}_0'\mathbf{A}\mathbf{v}_0| &= |\mathbf{v}^{*\prime}\mathbf{A}(\mathbf{v}^* - \mathbf{v}_0) + (\mathbf{v}^* - \mathbf{v}_0)'\mathbf{A}\mathbf{v}_0| \\ &\leqslant \|\mathbf{A}\|_{\mathrm{op}}\|\mathbf{v}^* - \mathbf{v}_0\| + \|\mathbf{v}^* - \mathbf{v}_0\|\|\mathbf{A}\|_{\mathrm{op}} \leqslant 2 \cdot \frac{1}{4} \cdot \|\mathbf{A}\|_{\mathrm{op}} = \frac{1}{2}\|\mathbf{A}\|_{\mathrm{op}}. \end{aligned}$$

Hence $\|\mathbf{A}\|_{\mathrm{op}} = |\mathbf{v}^{*\prime}\mathbf{A}\mathbf{v}^*| \leqslant |\mathbf{v}_0'\mathbf{A}\mathbf{v}_0| + \frac{1}{2}\|\mathbf{A}\|_{\mathrm{op}}$, which gives $\|\mathbf{A}\|_{\mathrm{op}} \leqslant 2|\mathbf{v}_0'\mathbf{A}\mathbf{v}_0| \leqslant 2\sup_{\mathbf{v} \in \mathcal{N}} |\mathbf{v}'\mathbf{A}\mathbf{v}|$. $\square$

6 Covariance Estimation

6.1 Setup

Remark. In the classical linear model, the design matrix $\mathbf{X}$ is treated as fixed and the Gram matrix $\mathbf{X}'\mathbf{X}$ is assumed to have full rank. In the modern setting, the rows $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^p$ of $\mathbf{X}$ are i.i.d. draws from a distribution with mean $\mathbf{0}$ and covariance $\mathbf{\Sigma} = \mathsf{E}[\mathbf{x}_i\mathbf{x}_i']$. The sample covariance is $\hat{\mathbf{\Sigma}} = \frac{1}{n}\mathbf{X}'\mathbf{X} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i'$, and the fundamental question is: *how close is $\hat{\mathbf{\Sigma}}$ to $\mathbf{\Sigma}$?*

Definition 6.1 (Isotropic Random Vector)**.** A random vector $\mathbf{x} \in \mathbb{R}^p$ is **isotropic** if $\mathsf{E}[\mathbf{x}] = \mathbf{0}$ and $\mathsf{E}[\mathbf{x}\mathbf{x}'] = \mathbf{I}_p$.

Remark. Any mean-zero random vector with invertible covariance $\mathbf{\Sigma}$ can be made isotropic by the transformation $\mathbf{z} = \mathbf{\Sigma}^{-1/2}\mathbf{x}$, since $\mathsf{E}[\mathbf{z}\mathbf{z}'] = \mathbf{\Sigma}^{-1/2}\mathbf{\Sigma}\mathbf{\Sigma}^{-1/2} = \mathbf{I}_p$. Working with isotropic vectors simplifies notation; all results extend to general $\mathbf{\Sigma}$ via this whitening transformation.

6.2 Operator Norm Bound for Covariance Estimation

Theorem 6.1 (Covariance Estimation for Sub-Gaussian Designs)**.** Let $\mathbf{x}_1, \ldots, \mathbf{x}_n$ be independent, isotropic, sub-Gaussian random vectors in $\mathbb{R}^p$ with $\|\mathbf{x}_i\|_{\psi_2} \leqslant K$. Let $\hat{\mathbf{\Sigma}} = \frac{1}{n}\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i'$. There exist universal constants $C, c > 0$ such that for every $\delta \in (0,1)$, if $n \geqslant CK^4 p/\delta^2$, then with probability at least $1 - 2\exp(-c\delta^2 n/K^4)$:

$$\boxed{\|\hat{\mathbf{\Sigma}} - \mathbf{I}_p\|_{\mathrm{op}} \leqslant \delta.} \tag{22}$$

Proof. **Step 1: Variational representation.** Since $\hat{\mathbf{\Sigma}} - \mathbf{I}_p$ is symmetric:

$$\|\hat{\mathbf{\Sigma}} - \mathbf{I}_p\|_{\mathrm{op}} = \sup_{\mathbf{v}\in S^{p-1}} |\mathbf{v}'(\hat{\mathbf{\Sigma}} - \mathbf{I}_p)\mathbf{v}| = \sup_{\mathbf{v}\in S^{p-1}} \left| \frac{1}{n}\sum_{i=1}^n \left[(\mathbf{v}'\mathbf{x}_i)^2 - 1\right] \right|.$$

Step 2: Pointwise concentration. Fix $\mathbf{v} \in S^{p-1}$. The random variable $Z_i = (\mathbf{v}'\mathbf{x}_i)^2 - 1$ has $\mathsf{E}[Z_i] = 0$ (isotropic assumption). Since $\mathbf{v}'\mathbf{x}_i$ is sub-Gaussian with $\|\mathbf{v}'\mathbf{x}_i\|_{\psi_2} \leqslant K$, by Proposition 4.1, $(\mathbf{v}'\mathbf{x}_i)^2$ is sub-exponential with $\|(\mathbf{v}'\mathbf{x}_i)^2\|_{\psi_1} \leqslant K^2$. Hence Z_i is sub-exponential with $\|Z_i\|_{\psi_1} \leqslant CK^2$. By Bernstein's inequality (Theorem 4.1):

$$\mathsf{P}\left(\left| \frac{1}{n}\sum_{i=1}^n Z_i \right| \geqslant t \right) \leqslant 2\exp\left[-cn \min\left(\frac{t^2}{K^4}, \frac{t}{K^2} \right) \right]. \tag{23}$$

For $t \leqslant 1$ (the regime of interest), $\min(t^2/K^4, t/K^2) = t^2/K^4$ when $t \leqslant K^2$, so the bound is $2\exp(-cnt^2/K^4)$.

Step 3: Discretization. Let $\mathcal{N}$ be a (1/4)-net of S^{p-1}. By Lemma 5.1, $|\mathcal{N}| \leqslant 9^p$. By Lemma 5.2, $\|\hat{\mathbf{\Sigma}} - \mathbf{I}_p\|_{\mathrm{op}} \leqslant 2\sup_{\mathbf{v}\in\mathcal{N}} |\mathbf{v}'(\hat{\mathbf{\Sigma}} - \mathbf{I}_p)\mathbf{v}|$.

Step 4: Union bound. Set $t = \delta/2$ in (23). For each $\mathbf{v} \in \mathcal{N}$:

$$\mathsf{P}\Big(|\mathbf{v}'(\hat{\mathbf{\Sigma}} - \mathbf{I}_p)\mathbf{v}| \geqslant \delta/2 \Big) \leqslant 2\exp\left(-\frac{cn\delta^2}{4K^4} \right).$$

Taking the union bound over $\mathcal{N}$:

$$\begin{aligned} \mathsf{P}\Big(\|\hat{\mathbf{\Sigma}} - \mathbf{I}_p\|_{\mathrm{op}} \geqslant \delta \Big) &\leqslant |\mathcal{N}| \cdot 2\exp\left(-\frac{cn\delta^2}{4K^4} \right) \leqslant 2 \cdot 9^p \cdot \exp\left(-\frac{cn\delta^2}{4K^4} \right) \\ &= 2\exp\left(p\ln 9 - \frac{cn\delta^2}{4K^4} \right). \end{aligned}$$

When $n \geqslant CK^4 p/\delta^2$ with $C = 8\ln 9/c$, the exponent is $\leqslant -cn\delta^2/(8K^4)$. Hence $\mathsf{P}(\|\hat{\mathbf{\Sigma}} - \mathbf{I}_p\|_{\mathrm{op}} \geqslant \delta) \leqslant 2\exp(-c'\delta^2 n/K^4)$ with $c' = c/8$. □

7 Eigenvalues of $\mathbf{X}'\mathbf{X}/n$: When Is the Design Well-Conditioned?

7.1 From Covariance Estimation to Eigenvalue Bounds

Theorem 7.1 (Eigenvalue Bounds for Sub-Gaussian Designs)**.** Under the conditions of Theorem 6.1, if $n \geqslant CK^4 p$, then with probability at least $1 - 2\exp(-cp)$:

$$\boxed{\frac{1}{2} \leqslant \lambda_{\min}\left(\frac{\mathbf{X}'\mathbf{X}}{n} \right) \leqslant \lambda_{\max}\left(\frac{\mathbf{X}'\mathbf{X}}{n} \right) \leqslant \frac{3}{2}.} \tag{24}$$

Equivalently, $\frac{n}{2}\mathbf{I}_p \preceq \mathbf{X}'\mathbf{X} \preceq \frac{3n}{2}\mathbf{I}_p$.

Proof. Apply Theorem 6.1 with $\delta = 1/2$. The sample-size condition becomes $n \geqslant 4CK^4p$. On the resulting high-probability event, $\|\mathbf{X}'\mathbf{X}/n - \mathbf{I}_p\|_{\mathrm{op}} \leqslant 1/2$. Since $\|\mathbf{A}\|_{\mathrm{op}} = \max(|\lambda_{\min}(\mathbf{A})|, |\lambda_{\max}(\mathbf{A})|)$ for symmetric $\mathbf{A}$, every eigenvalue λ_j of $\mathbf{X}'\mathbf{X}/n$ satisfies $|\lambda_j - 1| \leqslant 1/2$, giving $1/2 \leqslant \lambda_j \leqslant 3/2$. The probability bound is $1 - 2\exp(-c'n/(4K^4)) \geqslant 1 - 2\exp(-c'p)$ since $n \geqslant 4CK^4p$. □

Remark (Connection to the classical full-rank assumption)**.** Theorem 7.1 is the non-asymptotic replacement for the classical assumption (A4), $\mathrm{rank}(\mathbf{X}) = k + 1$. The classical assumption is qualitative (invertibility or not); the modern bound is quantitative: it says *how well-conditioned* $\mathbf{X}'\mathbf{X}$ is. Specifically, $\kappa(\mathbf{X}'\mathbf{X}/n) = \lambda_{\max}/\lambda_{\min} \leqslant 3$ with high probability whenever $n \gtrsim K^4p$. This is the sample-size condition under which OLS is numerically stable.

7.2 The Condition Number and Its Role

Definition 7.1 (Condition Number)**.** The **condition number** of a positive definite matrix $\mathbf{A}$ is

$$\kappa(\mathbf{A}) = \frac{\lambda_{\max}(\mathbf{A})}{\lambda_{\min}(\mathbf{A})} = \|\mathbf{A}\|_{\mathrm{op}} \cdot \|\mathbf{A}^{-1}\|_{\mathrm{op}}. \tag{25}$$

Proposition 7.1 (Condition Number of $\mathbf{X}'\mathbf{X}$ for General Covariance)**.** Let $\mathbf{x}_1, \ldots, \mathbf{x}_n$ be i.i.d. sub-Gaussian with covariance $\mathbf{\Sigma}$ (not necessarily $\mathbf{I}_p$). If $n \geqslant CK^4p$, then with high probability:

$$\kappa\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right) \leqslant 3\,\kappa(\mathbf{\Sigma}). \tag{26}$$

Proof. Apply Theorem 7.1 to the whitened vectors $\mathbf{z}_i = \mathbf{\Sigma}^{-1/2}\mathbf{x}_i$ (which are isotropic sub-Gaussian). Then $\frac{1}{n}\sum \mathbf{z}_i\mathbf{z}_i' = \mathbf{\Sigma}^{-1/2}\hat{\mathbf{\Sigma}}\mathbf{\Sigma}^{-1/2}$, and the eigenvalues of this matrix lie in $[1/2, 3/2]$. The eigenvalues of $\hat{\mathbf{\Sigma}} = \mathbf{X}'\mathbf{X}/n$ are those of $\mathbf{\Sigma}^{1/2}(\frac{1}{n}\sum \mathbf{z}_i\mathbf{z}_i')\mathbf{\Sigma}^{1/2}$. By the min-max theorem:

$$\lambda_{\max}(\hat{\mathbf{\Sigma}}) \leqslant \frac{3}{2}\lambda_{\max}(\mathbf{\Sigma}), \qquad \lambda_{\min}(\hat{\mathbf{\Sigma}}) \geqslant \frac{1}{2}\lambda_{\min}(\mathbf{\Sigma}).$$

Hence $\kappa(\hat{\mathbf{\Sigma}}) \leqslant (3/2)/(1/2) \cdot \kappa(\mathbf{\Sigma}) = 3\kappa(\mathbf{\Sigma})$. □

8 The Proportional Regime $p/n \to \gamma$

8.1 Motivation: The Marchenko–Pastur Law

Remark. When p is fixed and $n \to \infty$, the law of large numbers ensures $\hat{\mathbf{\Sigma}} \to \mathbf{\Sigma}$ and the design is always well-conditioned for large n. But in modern applications (genomics, finance, natural language), p can be comparable to or even larger than n. The **proportional regime** studies $p, n \to \infty$ with $p/n \to \gamma \in (0, \infty)$. The Marchenko–Pastur law describes the asymptotic eigenvalue distribution in this regime.

Theorem 8.1 (Marchenko–Pastur Law (Statement Only))**.** Let $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^p$ be i.i.d. with $\mathsf{E}[\mathbf{x}_i] = \mathbf{0}$ and $\mathsf{E}[\mathbf{x}_i\mathbf{x}_i'] = \mathbf{I}_p$, with i.i.d. entries having unit variance and finite fourth moment. Let $p/n \to \gamma \in (0, \infty)$. The empirical spectral distribution of $\hat{\mathbf{\Sigma}} = \frac{1}{n}\sum \mathbf{x}_i\mathbf{x}_i'$ converges weakly almost surely to the **Marchenko–Pastur distribution** F_γ with density

$$f_\gamma(x) = \frac{1}{2\pi\gamma x}\sqrt{(\lambda_+ - x)(x - \lambda_-)}\,\mathbf{1}_{[\lambda_-,\,\lambda_+]}(x), \tag{27}$$

where $\lambda_\pm = (1 \pm \sqrt{\gamma})^2$. When $\gamma > 1$, the distribution additionally has a point mass $(1 - 1/\gamma)$ at zero.

Remark (Implications for $\mathbf{X}'\mathbf{X}$)**.** The Marchenko–Pastur law reveals three qualitatively different regimes:

(1) **Classical regime** ($\gamma \to 0$, i.e., $p \ll n$): $\lambda_\pm \to 1$. The eigenvalues of $\hat{\mathbf{\Sigma}}$ cluster around 1. The design is well-conditioned; the classical theory applies.

(2) **Proportional regime** ($\gamma \in (0,1)$): $\lambda_- = (1-\sqrt{\gamma})^2 > 0$, $\lambda_+ = (1+\sqrt{\gamma})^2$. The design is invertible but the condition number $\kappa \approx \lambda_+/\lambda_- = ((1+\sqrt{\gamma})/(1-\sqrt{\gamma}))^2 \to \infty$ as $\gamma \to 1^-$. This is the "hard" regime where classical methods still work but their performance degrades.

(3) **Overparametrized regime** ($\gamma > 1$, i.e., $p > n$): $\lambda_- = 0$. The design matrix $\mathbf{X}$ has a nontrivial null space, $\mathbf{X}'\mathbf{X}$ is singular, and the OLS estimator $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ does not exist in the classical sense. Regularization (ridge, lasso) or the Moore–Penrose pseudoinverse is needed.

8.2 Non-Asymptotic Bounds in the Proportional Regime

Theorem 8.2 (Extreme Eigenvalue Bounds, Proportional Regime)**.** Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be independent, isotropic, sub-Gaussian random vectors in $\mathbb{R}^p$ with $\|\mathbf{x}_i\|_{\psi_2} \leqslant K$. There exist constants $C, c > 0$ depending only on K such that for every $t \geqslant 0$, with probability at least $1 - 2\exp(-ct^2)$:

$$\sqrt{n} - C\sqrt{p} - t \leqslant \sigma_{\min}(\mathbf{X}) \leqslant \sigma_{\max}(\mathbf{X}) \leqslant \sqrt{n} + C\sqrt{p} + t. \tag{28}$$

In terms of $\mathbf{X}'\mathbf{X}/n$:

$$\left(1 - C\sqrt{\frac{p}{n}} - \frac{t}{\sqrt{n}}\right)^2 \leqslant \lambda_{\min}\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right) \leqslant \lambda_{\max}\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right) \leqslant \left(1 + C\sqrt{\frac{p}{n}} + \frac{t}{\sqrt{n}}\right)^2. \tag{29}$$

Proof. **Upper bound on $\sigma_{\max}(\mathbf{X}) = \|\mathbf{X}\|_{\mathrm{op}}$.**

Step 1: Pointwise bound. For fixed $\mathbf{u} \in S^{p-1}$ and $\mathbf{v} \in S^{n-1}$, $\mathbf{u}'\mathbf{X}'\mathbf{v} = \sum_{i=1}^n v_i \langle \mathbf{x}_i, \mathbf{u}\rangle$. Since $\mathbf{x}_i$ is sub-Gaussian with $\|\langle \mathbf{x}_i, \mathbf{u}\rangle\|_{\psi_2} \leqslant K\|\mathbf{u}\| = K$, and $\|v_i\langle \mathbf{x}_i, \mathbf{u}\rangle\|_{\psi_2} \leqslant |v_i|K$, by Theorem 3.1:

$$\mathsf{P}(|\mathbf{u}'\mathbf{X}'\mathbf{v}| \geqslant s) \leqslant 2\exp\left(-\frac{cs^2}{K^2\|\mathbf{v}\|^2}\right) = 2\exp\left(-\frac{cs^2}{K^2}\right).$$

Step 2: ε-nets. Let $\mathcal{N}_p$ be a (1/4)-net of S^{p-1} and $\mathcal{N}_n$ a (1/4)-net of S^{n-1}, with $|\mathcal{N}_p| \leqslant 9^p$ and $|\mathcal{N}_n| \leqslant 9^n$ (Lemma 5.1). By the bilinear ε-net argument (as in Lemma 5.2, applied to the non-symmetric matrix $\mathbf{X}'$ via $\|\mathbf{X}\|_{\mathrm{op}} = \sup|\mathbf{u}'\mathbf{X}'\mathbf{v}|$): $\|\mathbf{X}\|_{\mathrm{op}} \leqslant 2\sup_{\mathbf{u}\in\mathcal{N}_p, \mathbf{v}\in\mathcal{N}_n}|\mathbf{u}'\mathbf{X}'\mathbf{v}|$.

Step 3: Union bound. Setting s in Step 1 and taking the union:

$$\mathsf{P}(\|\mathbf{X}\|_{\mathrm{op}} \geqslant 2s) \leqslant 2 \cdot 9^{p+n} \cdot \exp(-cs^2/K^2).$$

We need $cs^2/K^2 \geqslant (p+n)\ln 9 + \ln(2) + u^2$ for the tail probability $2e^{-u^2}$. Setting $s = CK(\sqrt{n} + \sqrt{p} + u)/2$ for C large enough: $s^2/K^2 \geqslant c'(n + p + u^2)$, and the 9^{p+n} factor is absorbed. Hence with probability $\geqslant 1 - 2e^{-cu^2}$:

$$\sigma_{\max}(\mathbf{X}) \leqslant CK(\sqrt{n} + \sqrt{p} + u).$$

Reparametrizing $t = CKu$ and absorbing CK into the constant C: $\sigma_{\max}(\mathbf{X}) \leqslant \sqrt{n} + C\sqrt{p} + t$.

Lower bound on $\sigma_{\min}(\mathbf{X}) = \inf_{\mathbf{u}\in S^{p-1}}\|\mathbf{X}\mathbf{u}\|$.

Step 4: Pointwise lower bound. Fix $\mathbf{u} \in S^{p-1}$. Then $\|\mathbf{X}\mathbf{u}\|^2 = \sum_{i=1}^n (\langle \mathbf{x}_i, \mathbf{u}\rangle)^2$. Each $Z_i = (\langle \mathbf{x}_i, \mathbf{u}\rangle)^2 - 1$ is mean-zero, sub-exponential with $\|Z_i\|_{\psi_1} \leqslant CK^2$. By Bernstein's inequality:

$$\mathsf{P}\big(\big|\|\mathbf{X}\mathbf{u}\|^2 - n\big| \geqslant s\big) \leqslant 2\exp\left[-c\min\left(\frac{s^2}{nK^4}, \frac{s}{K^2}\right)\right].$$

Setting $s = n/2$ (sub-Gaussian regime since $s^2/(nK^4) = n/(4K^4)$): $\mathsf{P}(\|\mathbf{X}\mathbf{u}\|^2 \leqslant n/2) \leqslant 2\exp(-cn/K^4)$.

Step 5: ε-net for infimum. Let $\mathcal{N}_p$ be a (1/4)-net of S^{p-1}. For any $\mathbf{u} \in S^{p-1}$, pick $\mathbf{u}_0 \in \mathcal{N}_p$ with $\|\mathbf{u} - \mathbf{u}_0\| \leqslant 1/4$. Then:

$$\|\mathbf{X}\mathbf{u}\| \geqslant \|\mathbf{X}\mathbf{u}_0\| - \|\mathbf{X}(\mathbf{u} - \mathbf{u}_0)\| \geqslant \|\mathbf{X}\mathbf{u}_0\| - \|\mathbf{X}\|_{\mathrm{op}}/4.$$

Hence $\sigma_{\min}(\mathbf{X}) \geqslant \min_{\mathbf{u}_0 \in \mathcal{N}_p} \|\mathbf{X}\mathbf{u}_0\| - \sigma_{\max}(\mathbf{X})/4$.

Step 6: Union bound. By Step 4 and the union over $|\mathcal{N}_p| \leqslant 9^p$:

$$\mathsf{P}\left(\min_{\mathcal{N}_p} \|\mathbf{X}\mathbf{u}_0\| \leqslant \sqrt{n/2}\right) \leqslant 9^p \cdot 2\exp(-cn/K^4).$$

When $n \geqslant C'K^4 p$: $9^p e^{-cn/K^4} \leqslant e^{-c'p}$, so $\min_{\mathcal{N}_p} \|\mathbf{X}\mathbf{u}_0\| \geqslant \sqrt{n/2}$ with probability $\geqslant 1 - 2e^{-c'p}$. On the intersection with the upper-bound event (probability $\geqslant 1 - 2e^{-ct^2} - 2e^{-c'p}$):

$$\begin{aligned}\sigma_{\min}(\mathbf{X}) &\geqslant \sqrt{n/2} - (\sqrt{n} + C\sqrt{p} + t)/4 \\ &= \sqrt{n}(\sqrt{1/2} - 1/4) - C\sqrt{p}/4 - t/4.\end{aligned}$$

Since $\sqrt{1/2} - 1/4 \approx 0.457 > 0$, after adjusting constants: $\sigma_{\min}(\mathbf{X}) \geqslant \sqrt{n} - C''\sqrt{p} - t$ with the stated probability. □

Remark. Setting $t = 0$ in (28), the singular values of $\mathbf{X}$ concentrate around $[\sqrt{n} - C\sqrt{p},\ \sqrt{n} + C\sqrt{p}]$, and those of $\mathbf{X}'\mathbf{X}/n$ around $[(1 - C\sqrt{p/n})^2,\ (1 + C\sqrt{p/n})^2]$. This is the non-asymptotic counterpart of the Marchenko–Pastur support $[\lambda_-, \lambda_+] = [(1 - \sqrt{\gamma})^2, (1 + \sqrt{\gamma})^2]$.

Part III
The Linear Model Redone

9 The Modern Setup

Definition 9.1 (The Linear Model, Modern Version)**.** The data follow

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \mathbf{X} \in \mathbb{R}^{n\times p}, \tag{30}$$

where p replaces $k+1$ for the total number of regressors. The following assumptions are imposed:

(M1) **Random design**: The rows $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ of $\mathbf{X}$ are i.i.d. random vectors with $\mathsf{E}[\mathbf{x}_i] = \mathbf{0}$ and $\mathsf{E}[\mathbf{x}_i\mathbf{x}_i'] = \boldsymbol{\Sigma}$, where $\boldsymbol{\Sigma} \succ 0$.

(M2) **Sub-Gaussian design**: Each $\mathbf{x}_i$ is sub-Gaussian with $\|\mathbf{x}_i\|_{\psi_2} \leqslant K$.

(M3) **Sub-Gaussian noise**: The errors $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d., mean-zero, sub-Gaussian with $\|\varepsilon_i\|_{\psi_2} \leqslant \sigma$, and independent of $\mathbf{X}$.

(M4) **Sample size**: $n \geqslant C_0 K^4 p$ for a sufficiently large universal constant C_0.

Remark (Comparison with the classical assumptions)**.** (1) (M1) replaces the fixed-design assumption. The design is now stochastic.

(2) (M2) replaces (A4): it is the sub-Gaussian analogue of the full-rank condition. By Theorem 7.1, (M2) and (M4) together imply $\mathbf{X}'\mathbf{X}$ is invertible with high probability.

(3) (M3) replaces (A5): sub-Gaussian errors instead of normal errors. The variance parameter $\sigma^2 = \mathrm{var}(\varepsilon_i)$ is bounded by $C\|\varepsilon_i\|_{\psi_2}^2$ (Proposition 1.2).

(4) (M4) is the modern analogue of $n > k+1$ in (A4). It is a quantitative sample-size condition.

10 OLS Prediction Error

Theorem 10.1 (Non-Asymptotic Bound on Prediction Error)**.** Under assumptions (M1)–(M4), with probability at least $1 - 2\exp(-cp)$:

$$\boxed{\frac{1}{n}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 \leqslant C\sigma^2\,\frac{p}{n}.} \tag{31}$$

Proof. **Step 1: Express the prediction error.** The OLS estimator is $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$ (exactly as in the classical derivation, Equation (2.6) in the classical notes). The prediction error is:

$$\frac{1}{n}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 = \frac{1}{n}\|\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}\|^2 = \frac{1}{n}\|\mathbf{H}\boldsymbol{\varepsilon}\|^2 = \frac{1}{n}\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon}, \tag{32}$$

where $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is the hat matrix.

Step 2: Conditional expectation. Conditional on $\mathbf{X}$ (and hence on $\mathbf{H}$), the errors $\varepsilon_1, \dots, \varepsilon_n$ are independent sub-Gaussian with mean zero and $\|\varepsilon_i\|_{\psi_2} \leqslant \sigma$. By the Hanson–Wright inequality (Theorem 3.2) applied conditionally on $\mathbf{X}$:

$$\mathsf{P}(|\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} - \mathsf{E}[\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \mid \mathbf{X}]| \geqslant t \mid \mathbf{X}) \leqslant 2\exp\left[-c\min\left(\frac{t^2}{\sigma^4\|\mathbf{H}\|_{\mathrm{F}}^2}, \frac{t}{\sigma^2\|\mathbf{H}\|_{\mathrm{op}}}\right)\right].$$

Since $\mathbf{H}$ is idempotent with $\mathrm{rank}(\mathbf{H}) = p$ (under the event that $\mathbf{X}$ has full rank):

- $\|\mathbf{H}\|_{\text{op}} = 1$ (eigenvalues are 0 and 1).
- $\|\mathbf{H}\|_{\text{F}}^2 = \text{tr}(\mathbf{H}) = p$.
- $\mathsf{E}[\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \mid \mathbf{X}] = \sigma_\varepsilon^2 \,\text{tr}(\mathbf{H}) = \sigma_\varepsilon^2 p$, where $\sigma_\varepsilon^2 = \text{var}(\varepsilon_i) \leqslant C\sigma^2$.

Here $\sigma_\varepsilon^2 = \mathsf{E}[\varepsilon_i^2]$ is the actual error variance. The Hanson–Wright bound gives, for $t = \delta\sigma^2 p$ with $\delta \in (0, 1)$:

$$\mathsf{P}\big(|\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} - \sigma_\varepsilon^2 p| \geqslant \delta\sigma^2 p \;\big|\; \mathbf{X}\big) \leqslant 2\exp\big[-c\min(\delta^2 p,\ \delta p)\big] \leqslant 2\exp(-c'\delta^2 p).$$

Step 3: Combine. Setting $\delta = 1$ (say), the event $\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \leqslant 2\sigma^2 p$ holds with probability $\geqslant 1 - 2\exp(-cp)$ conditional on $\mathbf{X}$, provided $\mathbf{X}$ has full rank. By Theorem 7.1, $\mathbf{X}$ has full rank with probability $\geqslant 1 - 2\exp(-cp)$ under (M4). By the union bound over both events:

$$\frac{1}{n}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 = \frac{1}{n}\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \leqslant \frac{2\sigma^2 p}{n} = C\sigma^2\frac{p}{n}$$

with probability $\geqslant 1 - 4\exp(-cp)$. □

Remark. The bound $\frac{1}{n}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 \lesssim \sigma^2 p/n$ says that the **in-sample prediction error** scales linearly with p/n. This is a non-asymptotic version of the well-known heuristic: "each estimated parameter costs you σ^2/n in prediction accuracy." In the classical theory, the exact result (under normality) is $\mathsf{E}[\frac{1}{n}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2] = \sigma^2 p/n$. The modern bound matches this rate but holds with high probability and under sub-Gaussian (not Gaussian) errors.

11 OLS Estimation Error

Theorem 11.1 (Non-Asymptotic Bound on Estimation Error)**.** Under assumptions (M1)–(M4), with probability at least $1 - C\exp(-cp)$:

$$\boxed{\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \leqslant \frac{C\sigma^2 p}{n\,\lambda_{\min}(\boldsymbol{\Sigma})}.} \tag{33}$$

More precisely, on the high-probability event from Theorem 7.1:

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'\mathbf{I}_p(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \leqslant \frac{2}{n\,\lambda_{\min}(\boldsymbol{\Sigma})}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 \leqslant \frac{C'\sigma^2 p}{n\,\lambda_{\min}(\boldsymbol{\Sigma})}. \tag{34}$$

Proof. **Step 1: Relate estimation error to prediction error.** For any $\mathbf{v} \in \mathbb{R}^p$:

$$\frac{1}{n}\|\mathbf{X}\mathbf{v}\|^2 = \mathbf{v}'\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)\mathbf{v} \geqslant \lambda_{\min}\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right)\|\mathbf{v}\|^2.$$

Hence $\|\mathbf{v}\|^2 \leqslant \frac{1}{\lambda_{\min}(\mathbf{X}'\mathbf{X}/n)} \cdot \frac{1}{n}\|\mathbf{X}\mathbf{v}\|^2$. Applying this to $\mathbf{v} = \hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$:

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \leqslant \frac{1}{\lambda_{\min}(\mathbf{X}'\mathbf{X}/n)} \cdot \frac{1}{n}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2. \tag{35}$$

Step 2: Bound $\lambda_{\min}(\mathbf{X}'\mathbf{X}/n)$ from below. By Theorem 7.1 applied to the whitened design $= \mathbf{X}\boldsymbol{\Sigma}^{-1/2}$, on the high-probability event:

$$\lambda_{\min}\left(\frac{\mathbf{X}'\mathbf{X}}{n}\right) \geqslant \frac{1}{2}\lambda_{\min}(\boldsymbol{\Sigma}).$$

Step 3: Combine. Substituting the prediction error bound (Theorem 10.1) and the eigenvalue bound into (35):

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \leqslant \frac{2}{\lambda_{\min}(\boldsymbol{\Sigma})} \cdot C\sigma^2\frac{p}{n} = \frac{C'\sigma^2 p}{n\,\lambda_{\min}(\boldsymbol{\Sigma})}.$$

The probability is at least $1 - C''\exp(-cp)$ by the union bound over the two high-probability events. □

Remark (Dependence on the condition number)**.** The estimation error bound (33) involves $1/\lambda_{\min}(\mathbf{\Sigma})$, which equals $\|\mathbf{\Sigma}^{-1}\|_{\mathrm{op}}$. When the population covariance is well-conditioned ($\kappa(\mathbf{\Sigma})$ small), $\lambda_{\min}(\mathbf{\Sigma}) \approx \lambda_{\max}(\mathbf{\Sigma}) \approx \|\mathbf{\Sigma}\|_{\mathrm{op}}$, and the bound is comparable to $\sigma^2 p/(n\|\mathbf{\Sigma}\|_{\mathrm{op}})$. But when $\mathbf{\Sigma}$ is ill-conditioned (e.g., near-multicollinearity), $\lambda_{\min}(\mathbf{\Sigma}) \to 0$ and the estimation error explodes. This is the modern analogue of the classical observation that near-multicollinearity inflates $\mathrm{var}(\hat{\beta}_j) = \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$.

12 Concentration of the Residual Sum of Squares

In the classical theory, $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-p}$ under normality (Theorem 4.5 in the classical notes). The following theorem is its non-asymptotic, sub-Gaussian counterpart.

Theorem 12.1 (Concentration of SSE)**.** Under assumptions (M1)–(M4), with probability at least $1 - C\exp(-cp)$:

$$\left|\frac{\mathrm{SSE}}{\sigma_\varepsilon^2} - (n-p)\right| \leqslant C'\sqrt{n-p}, \tag{36}$$

where $\sigma_\varepsilon^2 = \mathrm{var}(\varepsilon_i)$. Consequently, $\mathrm{MSE} = \mathrm{SSE}/(n-p)$ satisfies $|\mathrm{MSE} - \sigma_\varepsilon^2| \leqslant C'\sigma_\varepsilon^2/\sqrt{n-p}$.

Proof. **Step 1: Express** SSE **as a quadratic form.** We have $\mathrm{SSE} = \mathbf{e}'\mathbf{e} = (\mathbf{M}\boldsymbol{\varepsilon})'(\mathbf{M}\boldsymbol{\varepsilon}) = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}$, where $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$ is the residual maker. On the event that $\mathbf{X}$ has full column rank (which holds with probability $\geqslant 1 - 2\exp(-c_0 p)$ by Theorem 7.1), $\mathbf{M}$ is symmetric idempotent with $\mathrm{tr}(\mathbf{M}) = n-p$, $\|\mathbf{M}\|_{\mathrm{op}} = 1$, and $\|\mathbf{M}\|_{\mathrm{F}}^2 = \mathrm{tr}(\mathbf{M}) = n-p$.

Step 2: Conditional expectation. By the quadratic form expectation lemma (Lemma 3.7 of the classical notes), with $\mathsf{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$ and $\mathrm{var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma_\varepsilon^2\mathbf{I}_n$:

$$\mathsf{E}[\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon} \mid \mathbf{X}] = \sigma_\varepsilon^2\,\mathrm{tr}(\mathbf{M}) + \mathbf{0}'\mathbf{M}\mathbf{0} = \sigma_\varepsilon^2(n-p).$$

Step 3: Concentration via Hanson–Wright. Conditional on $\mathbf{X}$, the vector $\boldsymbol{\varepsilon}$ has independent, mean-zero, sub-Gaussian components with $\|\varepsilon_i\|_{\psi_2} \leqslant \sigma$. By the Hanson–Wright inequality (Theorem 3.2):

$$\mathsf{P}\big(|\boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon} - \sigma_\varepsilon^2(n-p)| \geqslant t \mid \mathbf{X}\big) \leqslant 2\exp\left[-c\min\left(\frac{t^2}{\sigma^4(n-p)}, \frac{t}{\sigma^2}\right)\right].$$

Here we used $\|\mathbf{M}\|_{\mathrm{F}}^2 = n-p$ and $\|\mathbf{M}\|_{\mathrm{op}} = 1$.

Step 4: Choose t**.** Set $t = \delta\sigma_\varepsilon^2(n-p)$ with $\delta > 0$. The first argument of the minimum is $\delta^2\sigma_\varepsilon^4(n-p)/\sigma^4$, and the second is $\delta\sigma_\varepsilon^2(n-p)/\sigma^2$. Since $\sigma_\varepsilon^2 \leqslant C\sigma^2$ (Proposition 1.2) and $\delta \leqslant 1$, both arguments are $\geqslant c'\delta^2(n-p)$ for a constant c' depending on C. Hence:

$$\mathsf{P}\left(\left|\frac{\mathrm{SSE}}{\sigma_\varepsilon^2} - (n-p)\right| \geqslant \delta(n-p) \;\middle|\; \mathbf{X}\right) \leqslant 2\exp(-c''\delta^2(n-p)).$$

Choosing $\delta = C_0/\sqrt{n-p}$ for a constant C_0:

$$\left|\frac{\mathrm{SSE}}{\sigma_\varepsilon^2} - (n-p)\right| \leqslant C_0\sqrt{n-p}$$

with conditional probability $\geqslant 1 - 2\exp(-c''C_0^2)$.

Step 5: Remove conditioning. By the union bound over the full-rank event and the concentration event:

$$\mathsf{P}\left(\left|\frac{\mathrm{SSE}}{\sigma_\varepsilon^2} - (n-p)\right| \leqslant C_0\sqrt{n-p}\right) \geqslant 1 - 2\exp(-c''C_0^2) - 2\exp(-c_0 p).$$

Choosing C_0 large enough and noting $n - p \geqslant c_1 p$ under (M4): the probability is $\geqslant 1 - C\exp(-cp)$. Dividing by $n-p$: $|\mathrm{MSE} - \sigma_\varepsilon^2| \leqslant C_0\sigma_\varepsilon^2/\sqrt{n-p}$. □

13 Non-Asymptotic Confidence Intervals Without Normality

Theorem 13.1 (Sub-Gaussian Confidence Interval for β_j)**.** Under assumptions (M1)–(M4), for each $j \in \{1, \dots, p\}$ and $\alpha \in (0, 1)$: conditionally on $\mathbf{X}$ having full rank, with probability at least $1 - \alpha$,

$$|\hat{\beta}_j - \beta_j| \leqslant C\sigma\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj} \cdot \ln \frac{2}{\alpha}}. \tag{37}$$

Proof. **Step 1: Isolate $\hat{\beta}_j - \beta_j$.** Let $\mathbf{e}_j \in \mathbb{R}^p$ be the j-th standard basis vector. Then:

$$\hat{\beta}_j - \beta_j = \mathbf{e}_j'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \mathbf{e}_j'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} = \mathbf{c}'\boldsymbol{\varepsilon},$$

where $\mathbf{c} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{e}_j \in \mathbb{R}^n$.

Step 2: Conditional sub-Gaussian bound. Fix $\mathbf{X}$ (hence $\mathbf{c}$ is fixed). Then $\mathbf{c}'\boldsymbol{\varepsilon} = \sum_{i=1}^n c_i\varepsilon_i$ is a sum of independent, mean-zero, sub-Gaussian random variables with $\|c_i\varepsilon_i\|_{\psi_2} \leqslant |c_i|\sigma$. By Theorem 3.1:

$$\mathsf{P}(|\mathbf{c}'\boldsymbol{\varepsilon}| \geqslant t \mid \mathbf{X}) \leqslant 2\exp\left(-\frac{ct^2}{\sigma^2\|\mathbf{c}\|^2}\right).$$

Step 3: Compute $\|\mathbf{c}\|^2$.

$$\|\mathbf{c}\|^2 = \mathbf{c}'\mathbf{c} = \mathbf{e}_j'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{e}_j = \mathbf{e}_j'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{e}_j = [(\mathbf{X}'\mathbf{X})^{-1}]_{jj}.$$

Step 4: Invert the tail bound. Setting $2\exp(-ct^2/(\sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj})) = \alpha$ and solving:

$$t = \sigma\sqrt{\frac{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}{c} \cdot \ln \frac{2}{\alpha}} = C'\sigma\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj} \cdot \ln \frac{2}{\alpha}},$$

where $C' = 1/\sqrt{c}$. Hence $|\hat{\beta}_j - \beta_j| \leqslant t$ with probability $\geqslant 1 - \alpha$, conditional on $\mathbf{X}$. □

Remark (Comparison with the classical interval)**.** The classical t-based confidence interval (Theorem 6.1 of the classical notes) has half-width $t_{\alpha/2,n-p} \cdot s\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$, where $s = \sqrt{\text{MSE}}$. For large $n - p$, the quantile $t_{\alpha/2,n-p} \approx z_{\alpha/2}$, and the Gaussian tail approximation gives $z_{\alpha/2} \approx \sqrt{2\ln(2/\alpha)}$. So the modern and classical widths agree up to a constant factor. The modern interval has two advantages: (1) no normality assumption on $\boldsymbol{\varepsilon}$, and (2) finite-sample validity. Its disadvantage is that it uses the unknown σ rather than the data-driven s. This can be remedied: by Theorem 12.1, $s^2 = \text{MSE}$ satisfies $|s^2/\sigma_\varepsilon^2 - 1| \leqslant C/\sqrt{n-p}$ with high probability, so for large $n - p$ one may replace σ by $s\sqrt{2}$ (say) at the cost of doubling the constant and adding the failure probability of the MSE concentration event.

14 Comparison: Classical vs. Modern

Remark (Side-by-side comparison)**.** The following table compares the classical and modern treatments of the linear model. The notation $p = k + 1$.

Aspect	Classical Theory	Modern Theory
Design	Fixed ($\mathbf{X}$ non-random)	Random, i.i.d. rows
Error distribution	$\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$	Sub-Gaussian, $\|\varepsilon_i\|_{\psi_2} \leqslant \sigma$
Invertibility	$\operatorname{rank}(\mathbf{X}) = p$ (assumed)	$n \geqslant CK^4 p$ (implies w.h.p.)
Prediction error	$\mathsf{E}[\|\mathbf{H}\boldsymbol{\varepsilon}\|^2/n] = \sigma^2 p/n$ (exact)	$\|\mathbf{H}\boldsymbol{\varepsilon}\|^2/n \leqslant C\sigma^2 p/n$ (w.h.p.)
Estimation error	$\operatorname{var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$	$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \leqslant C\sigma^2 p/(n\lambda_{\min}(\boldsymbol{\Sigma}))$ (w.h.p.)
Distribution of SSE	$\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-p}$	$\lvert\mathrm{SSE}/\sigma^2 - (n-p)\rvert \leqslant C\sqrt{n-p}$ (w.h.p.)
Inference	t, F, χ^2 pivots	Sub-Gaussian concentration (Thm. 13.1)
Regime	p fixed, $n \to \infty$	$p/n \to \gamma \in (0,1)$ allowed

15 What Happens When p Is Comparable to n

Remark. As $p/n \to \gamma \in (0,1)$, the OLS estimator remains well-defined but its behavior changes qualitatively.

Theorem 15.1 (Prediction and Estimation Error in the Proportional Regime)**.** Under assumptions (M1)–(M3) with $\boldsymbol{\Sigma} = \mathbf{I}_p$ (isotropic design), conditional on $\mathbf{X}$ having full column rank:

(i) **In-sample prediction error**: $\mathsf{E}[\frac{1}{n}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 \mid \mathbf{X}] = \sigma_\varepsilon^2\, p/n$, where $\sigma_\varepsilon^2 = \operatorname{var}(\varepsilon_i)$.

(ii) **Unbiasedness of MSE**: $\mathsf{E}[\mathrm{MSE} \mid \mathbf{X}] = \sigma_\varepsilon^2$.

(iii) **Out-of-sample prediction risk**: For a new $\mathbf{x}_0$ drawn independently from the same distribution,

$$\mathsf{E}[(\mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}))^2 \mid \mathbf{X}] = \sigma_\varepsilon^2 \operatorname{tr}((\mathbf{X}'\mathbf{X})^{-1}). \tag{38}$$

As $p/n \to \gamma \in (0,1)$ with i.i.d. entries having finite fourth moment, $\frac{1}{p}\operatorname{tr}((\mathbf{X}'\mathbf{X}/n)^{-1}) \to \frac{1}{1-\gamma}$ almost surely by the Marchenko–Pastur law (Theorem 8.1), giving $\mathsf{E}[(\mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}))^2 \mid \mathbf{X}] \to \sigma_\varepsilon^2 \cdot \frac{\gamma}{1-\gamma}$.

Proof. **(i).** Conditional on $\mathbf{X}$: $\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 = \boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon}$ (from (32)). By the quadratic form expectation lemma (Lemma 3.7 of the classical notes) with $\mathsf{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$ and $\operatorname{var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma_\varepsilon^2 \mathbf{I}_n$:

$$\mathsf{E}[\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \mid \mathbf{X}] = \sigma_\varepsilon^2 \operatorname{tr}(\mathbf{H}) + \mathbf{0}'\mathbf{H}\mathbf{0} = \sigma_\varepsilon^2 p.$$

Dividing by n gives (i).

(ii). Similarly, $\mathrm{SSE} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}$ where $\mathbf{M} = \mathbf{I} - \mathbf{H}$ has $\operatorname{tr}(\mathbf{M}) = n - p$. So $\mathsf{E}[\mathrm{SSE} \mid \mathbf{X}] = \sigma_\varepsilon^2(n-p)$ and $\mathsf{E}[\mathrm{MSE} \mid \mathbf{X}] = \sigma_\varepsilon^2$.

(iii). Since $\mathbf{x}_0$ is independent of $(\mathbf{X}, \boldsymbol{\varepsilon})$ with $\mathsf{E}[\mathbf{x}_0\mathbf{x}_0'] = \mathbf{I}_p$:

$$\mathsf{E}[(\mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}))^2 \mid \mathbf{X}, \boldsymbol{\varepsilon}] = (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'\mathsf{E}[\mathbf{x}_0\mathbf{x}_0'](\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2.$$

Taking expectation over $\boldsymbol{\varepsilon}$ conditional on $\mathbf{X}$:

$$\begin{aligned}\mathsf{E}[\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \mid \mathbf{X}] &= \mathsf{E}[\boldsymbol{\varepsilon}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\boldsymbol{\varepsilon} \mid \mathbf{X}] \\ &= \mathsf{E}[\boldsymbol{\varepsilon}'(\mathbf{X}'\mathbf{X})^{-1}\boldsymbol{\varepsilon} \mid \mathbf{X}] \cdot \frac{\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}}{}\end{aligned}$$

More directly: $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$, so

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \boldsymbol{\varepsilon}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-2}\mathbf{X}'\boldsymbol{\varepsilon}.$$

Let $\mathbf{C} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-2}\mathbf{X}'$. By the quadratic form lemma: $\mathsf{E}[\boldsymbol{\varepsilon}'\mathbf{C}\boldsymbol{\varepsilon} \mid \mathbf{X}] = \sigma_\varepsilon^2 \operatorname{tr}(\mathbf{C})$. Using $\operatorname{tr}(\mathbf{AC}) = \operatorname{tr}(\mathbf{CA})$:

$$\operatorname{tr}(\mathbf{C}) = \operatorname{tr}(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-2}\mathbf{X}') = \operatorname{tr}((\mathbf{X}'\mathbf{X})^{-2}\mathbf{X}'\mathbf{X}) = \operatorname{tr}((\mathbf{X}'\mathbf{X})^{-1}).$$

Hence $\mathsf{E}[(\mathbf{x}_0'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}))^2 \mid \mathbf{X}] = \sigma_\varepsilon^2 \operatorname{tr}((\mathbf{X}'\mathbf{X})^{-1})$.

For the asymptotic statement: $\operatorname{tr}((\mathbf{X}'\mathbf{X})^{-1}) = n \cdot \frac{1}{n}\sum_{j=1}^p \frac{1}{\lambda_j(\mathbf{X}'\mathbf{X}/n)\cdot n} = \frac{1}{n}\sum_{j=1}^p \frac{1}{\lambda_j(\mathbf{X}'\mathbf{X}/n)}$. Wait: $(\mathbf{X}'\mathbf{X})^{-1}$ has eigenvalues $1/\lambda_j(\mathbf{X}'\mathbf{X})$, so $\operatorname{tr}((\mathbf{X}'\mathbf{X})^{-1}) = \sum_{j=1}^p 1/\lambda_j(\mathbf{X}'\mathbf{X}) = \frac{1}{n}\sum_{j=1}^p 1/\lambda_j(\mathbf{X}'\mathbf{X}/n)$. By the Marchenko–Pastur law, the empirical distribution of the eigenvalues $\lambda_j(\mathbf{X}'\mathbf{X}/n)$ converges to F_γ, so:

$$\frac{1}{p}\sum_{j=1}^p \frac{1}{\lambda_j(\mathbf{X}'\mathbf{X}/n)} \to \int \frac{1}{x}\,\mathrm{d}F_\gamma(x) = \frac{1}{1-\gamma},$$

where the last equality is the standard Marchenko–Pastur moment $\int x^{-1}\,\mathrm{d}F_\gamma(x) = 1/(1-\gamma)$ for $\gamma < 1$ (computed via the Stieltjes transform $m(z) = \int (x-z)^{-1}\mathrm{d}F_\gamma(x) = \frac{1-\gamma-z+\sqrt{(z-\lambda_-)(\lambda_+-z)}}{2\gamma z}$ evaluated at $z = 0$). Hence:

$$\sigma_\varepsilon^2 \operatorname{tr}((\mathbf{X}'\mathbf{X})^{-1}) = \sigma_\varepsilon^2 \sum_{j=1}^p \frac{1}{\lambda_j(\mathbf{X}'\mathbf{X})} = \sigma_\varepsilon^2 \cdot \frac{p}{n} \cdot \frac{1}{p}\sum_{j=1}^p \frac{1}{\lambda_j(\mathbf{X}'\mathbf{X}/n)} \to \sigma_\varepsilon^2 \cdot \gamma \cdot \frac{1}{1-\gamma} = \sigma_\varepsilon^2 \cdot \frac{\gamma}{1-\gamma}. \qquad \square$$

Corollary 15.1 (Estimation Error Diverges as $\gamma \to 1^-$)**.** Under the conditions of Theorem 15.1, $\mathsf{E}[\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \mid \mathbf{X}] = \sigma_\varepsilon^2 \operatorname{tr}((\mathbf{X}'\mathbf{X})^{-1}) \to \infty$ as $p/n \to 1^-$, since $\frac{1}{1-\gamma} \to \infty$.

Remark (The overparametrized regime $p > n$)**.** When $p > n$, $\mathbf{X}'\mathbf{X}$ is singular and the classical OLS estimator is undefined. The minimum-norm interpolator $\hat{\boldsymbol{\beta}}_{\mathrm{mn}} = \mathbf{X}'(\mathbf{X}\mathbf{X}')^{-1}\mathbf{Y}$ can still be computed, and—surprisingly—its prediction risk can decrease as p/n increases beyond 1. This "double descent" phenomenon is the starting point of modern interpolation theory (Hastie, Montanari, Rosset, and Tibshirani, 2022), but lies beyond the scope of these notes.

16 Exercises

Tier A: Reproduce the Derivation

Exercise 16.1. Let X be a random variable with $\mathsf{E}[X] = 0$ and $a \leqslant X \leqslant b$. Prove Hoeffding's lemma: $\mathsf{E}[e^{tX}] \leqslant e^{t^2(b-a)^2/8}$.

Exercise 16.2. Let $X_1, \ldots, X_n$ be independent with $\mathsf{E}[X_i] = 0$ and $a_i \leqslant X_i \leqslant b_i$. Prove Hoeffding's inequality: $\mathsf{P}(|\sum X_i| \geqslant t) \leqslant 2\exp(-2t^2/\sum(b_i - a_i)^2)$.

Exercise 16.3. Prove that if X is sub-Gaussian, then X^2 is sub-exponential with $\|X^2\|_{\psi_1} \leqslant C\|X\|_{\psi_2}^2$.

Exercise 16.4. Let $\mathbf{x}_1, \ldots, \mathbf{x}_n$ be i.i.d. isotropic sub-Gaussian in $\mathbb{R}^p$. Carry out the pointwise Bernstein argument (Step 2 of Theorem 6.1) to show that for a fixed $\mathbf{v} \in S^{p-1}$:

$$\mathsf{P}\left(\left|\frac{1}{n}\sum_{i=1}^n [(\mathbf{v}'\mathbf{x}_i)^2 - 1]\right| \geqslant t\right) \leqslant 2\exp\left[-cn\min\left(\frac{t^2}{K^4}, \frac{t}{K^2}\right)\right].$$

Exercise 16.5. Derive the prediction error identity $\frac{1}{n}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 = \frac{1}{n}\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon}$ and compute $\mathsf{E}[\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \mid \mathbf{X}]$ under sub-Gaussian errors.

Tier B: Apply the Theory

Exercise 16.6. Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli(p) with $p \in (0,1)$. Use Hoeffding's inequality to find the smallest n guaranteeing $\mathsf{P}(|\bar{X} - p| \geqslant 0.01) \leqslant 0.05$.

Exercise 16.7. Suppose $\mathbf{x}_i \sim N(\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} = \mathrm{diag}(1, 2, 3, \ldots, p)$. (a) Compute $\kappa(\boldsymbol{\Sigma})$. (b) How large must n be (in terms of p) to guarantee $\|\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_{\mathrm{op}}/\|\boldsymbol{\Sigma}\|_{\mathrm{op}} \leqslant 1/2$ with probability $\geqslant 0.99$? (c) Compare the estimation error bound from Theorem 11.1 with the classical variance formula $\sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{11}$.

Exercise 16.8. In the Marchenko–Pastur law with $\gamma = 0.5$, compute $\lambda_- = (1 - \sqrt{0.5})^2$ and $\lambda_+ = (1 + \sqrt{0.5})^2$. What is the asymptotic condition number of $\mathbf{X}'\mathbf{X}/n$? If the true error variance is $\sigma^2 = 1$, estimate $\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2$ using Corollary 15.1.

Exercise 16.9. A design matrix $\mathbf{X} \in \mathbb{R}^{100 \times 10}$ has i.i.d. Rademacher entries (± 1 with equal probability). (a) Verify that each row is sub-Gaussian and compute K. (b) Check the sample-size condition $n \geqslant CK^4 p$. (c) Give a high-probability interval for $\lambda_{\min}(\mathbf{X}'\mathbf{X}/n)$ and $\lambda_{\max}(\mathbf{X}'\mathbf{X}/n)$ using Theorem 8.2.

Tier C: Extend and Connect

Exercise 16.10. (Concentration of the MSE.) Under (M1)–(M4), the classical notes show $\mathsf{E}[\mathrm{MSE}] = \sigma^2$ under normality. Use the Hanson–Wright inequality to prove that $|\mathrm{SSE} - \sigma^2(n-p)| \leqslant C\sigma^2\sqrt{n-p}$ with high probability under sub-Gaussian errors. Conclude that $\mathrm{MSE} = \mathrm{SSE}/(n-p)$ concentrates around σ^2.

Exercise 16.11. (Ridge regression in the proportional regime.) Consider the ridge estimator $\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I}_p)^{-1}\mathbf{X}'\mathbf{Y}$. (a) Show that $\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I}_p)^{-1}(\mathbf{X}'\boldsymbol{\varepsilon} - \lambda\boldsymbol{\beta})$. (b) Decompose $\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2$ into a bias term (depending on $\lambda\boldsymbol{\beta}$) and a variance term (depending on $\boldsymbol{\varepsilon}$). (c) Why does ridge help when $p/n \to 1^-$?

Exercise 16.12. (Sub-Gaussian tail optimality.) Show that the Hoeffding bound for $\bar{X}$ (Corollary 2.1) cannot be improved beyond constant factors for Rademacher variables: explicitly compute $\mathsf{P}(\sum_{i=1}^n \varepsilon_i \geqslant t)$ for ε_i i.i.d. uniform on $\{-1, +1\}$ via the MGF, and compare with (6).

Exercise 16.13. (From concentration to confidence intervals.) Under (M1)–(M4), construct a non-asymptotic confidence interval for a single coefficient β_j *without assuming normality*. Specifically: (a) Show $\hat{\beta}_j - \beta_j = \mathbf{e}_j'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$, where $\mathbf{e}_j$ is the j-th standard basis vector. (b) Conditionally on $\mathbf{X}$, bound $|\hat{\beta}_j - \beta_j|$ using the sub-Gaussian tail of $\mathbf{c}'\boldsymbol{\varepsilon}$ for $\mathbf{c} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}_j$. (c) Show this yields a confidence interval of width $O(\sigma\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}\log(1/\alpha)})$ at level $1 - \alpha$. Compare with the classical interval width $t_{\alpha/2, n-p} \cdot s\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$.

17 Solutions to Exercises

Solutions to Tier A

Solution to Exercise 16.1. See the proof of Lemma 2.1.

Solution to Exercise 16.2. See the proof of Theorem 2.1.

Solution to Exercise 16.3. See the proof of Proposition 4.1.

Solution to Exercise 16.4. The random variable $Z_i = (\mathbf{v}'\mathbf{x}_i)^2 - 1$ satisfies $\mathsf{E}[Z_i] = 0$ (isotropic assumption) and is sub-exponential with $\|Z_i\|_{\psi_1} \leqslant CK^2$ (since $\mathbf{v}'\mathbf{x}_i$ is sub-Gaussian with norm $\leqslant K$ and squares of sub-Gaussians are sub-exponential by Proposition 4.1). By Bernstein's inequality (Theorem 4.1) applied to $\sum_{i=1}^n Z_i$:

$$\mathsf{P}\left(\left|\sum_{i=1}^n Z_i\right| \geqslant nt\right) \leqslant 2\exp\left[-c\min\left(\frac{n^2t^2}{nC^2K^4}, \frac{nt}{CK^2}\right)\right] = 2\exp\left[-cn\min\left(\frac{t^2}{K^4}, \frac{t}{K^2}\right)\right].$$

Solution to Exercise 16.5. Substituting $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$:

$$\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} = \mathbf{H}\boldsymbol{\varepsilon}.$$

Hence $\frac{1}{n}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 = \frac{1}{n}\|\mathbf{H}\boldsymbol{\varepsilon}\|^2 = \frac{1}{n}(\mathbf{H}\boldsymbol{\varepsilon})'\mathbf{H}\boldsymbol{\varepsilon} = \frac{1}{n}\boldsymbol{\varepsilon}'\mathbf{H}'\mathbf{H}\boldsymbol{\varepsilon} = \frac{1}{n}\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon}$ (using $\mathbf{H}' = \mathbf{H}$ and $\mathbf{H}^2 = \mathbf{H}$). For the conditional expectation: since $\varepsilon_1, \ldots, \varepsilon_n$ are independent with $\mathsf{E}[\varepsilon_i] = 0$ and $\mathsf{E}[\varepsilon_i^2] = \sigma_\varepsilon^2$, we have $\mathsf{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \mathbf{X}] = \sigma_\varepsilon^2\mathbf{I}_n$. By Lemma 3.7 of the classical notes (expected value of quadratic form, with $\boldsymbol{\mu} = \mathbf{0}$):

$$\mathsf{E}[\boldsymbol{\varepsilon}'\mathbf{H}\boldsymbol{\varepsilon} \mid \mathbf{X}] = \operatorname{tr}(\mathbf{H} \cdot \sigma_\varepsilon^2\mathbf{I}_n) = \sigma_\varepsilon^2 \operatorname{tr}(\mathbf{H}) = \sigma_\varepsilon^2 p.$$

Dividing by n: $\mathsf{E}[\frac{1}{n}\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 \mid \mathbf{X}] = \sigma_\varepsilon^2 p/n$.

Solutions to Tier B

Solution to Exercise 16.6. Each $X_i \in [0, 1]$, so $b_i - a_i = 1$. By Corollary 2.1: $\mathsf{P}(|\bar{X} - p| \geqslant 0.01) \leqslant 2\exp(-2n(0.01)^2) = 2e^{-n/5000}$. Setting $2e^{-n/5000} \leqslant 0.05$ gives $n \geqslant 5000\ln(40) \approx 18\,444$.

Solution to Exercise 16.7. (a) $\kappa(\boldsymbol{\Sigma}) = \lambda_{\max}/\lambda_{\min} = p/1 = p$. (b) The whitened vectors $\mathbf{z}_i = \boldsymbol{\Sigma}^{-1/2}\mathbf{x}_i$ are isotropic Gaussian with $\|\mathbf{z}_i\|_{\psi_2} \leqslant C$. The condition $\|\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_{\mathrm{op}}/\|\boldsymbol{\Sigma}\|_{\mathrm{op}} \leqslant 1/2$ reduces (after whitening) to $\|\hat{\mathbf{I}} - \mathbf{I}_p\|_{\mathrm{op}} \leqslant 1/2$ for the whitened sample covariance. By Theorem 6.1 with $\delta = 1/2$, we need $n \geqslant 4Cp$. For probability 0.99, we additionally need $2\exp(-cp) \leqslant 0.01$, i.e., $p \geqslant \frac{1}{c}\ln 200$, which is mild.

(c) The classical formula gives $\operatorname{var}(\hat{\beta}_1) = \sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{11}$, which depends on the specific realization of $\mathbf{X}$. For the diagonal covariance, $\mathsf{E}[(\mathbf{X}'\mathbf{X})^{-1}] \approx \frac{1}{n}\boldsymbol{\Sigma}^{-1}$ for large n, so $\operatorname{var}(\hat{\beta}_1) \approx \sigma^2/(n \cdot 1) = \sigma^2/n$. The modern bound gives $\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \leqslant C\sigma^2 p/(n\lambda_{\min}(\boldsymbol{\Sigma})) = C\sigma^2 p/n$, which is larger by a factor p (but bounds the total ℓ^2 error, not just one coordinate).

Solution to Exercise 16.8. $\lambda_- = (1 - 1/\sqrt{2})^2 = (1 - \sqrt{2}/2)^2 = (2 - \sqrt{2})^2/4 \approx 0.0858$. $\lambda_+ = (1 + \sqrt{2}/2)^2 = (2 + \sqrt{2})^2/4 \approx 2.914$. Asymptotic condition number: $\kappa \approx \lambda_+/\lambda_- \approx 33.97$. Estimation error: $\gamma/(1-\gamma) = 0.5/0.5 = 1$, so $\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \approx \sigma^2/(\lambda_{\min}(\boldsymbol{\Sigma})) \cdot 1 = 1/\lambda_{\min}(\boldsymbol{\Sigma})$ when $\sigma^2 = 1$ and $\boldsymbol{\Sigma} = \mathbf{I}_p$.

Solution to Exercise 16.9. (a) Each entry X_{ij} is Rademacher: $\|X_{ij}\|_{\psi_2} = C$ (by Example 1.1(2)). A row $\mathbf{x}_i = (X_{i1}, \ldots, X_{ip})'$ has $\|\langle \mathbf{x}_i, \mathbf{v}\rangle\|_{\psi_2} = \|\sum_j v_j X_{ij}\|_{\psi_2} \leqslant C\|\mathbf{v}\|$ by Proposition 1.1. So $K = C$. (b) $n = 100$, $p = 10$, $K^4 p = C^4 \cdot 10$. For reasonable C, $n = 100 \gg C^4 \cdot 10$, so the condition is satisfied. (c) By (29) with $t = 0$: $\lambda_{\min}(\mathbf{X}'\mathbf{X}/n) \gtrsim (1 - C'\sqrt{10/100})^2 = (1 - C'\sqrt{0.1})^2$ and $\lambda_{\max}(\mathbf{X}'\mathbf{X}/n) \lesssim (1 + C'\sqrt{0.1})^2$. With $C' \approx 1$: $\lambda_{\min} \gtrsim (1 - 0.316)^2 \approx 0.467$, $\lambda_{\max} \lesssim (1 + 0.316)^2 \approx 1.732$.

Solutions to Tier C

Solution to Exercise 16.10. We have $\mathrm{SSE} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}$ where $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$ is symmetric idempotent with $\operatorname{tr}(\mathbf{M}) = n - p$ and $\|\mathbf{M}\|_{\mathrm{op}} = 1$, $\|\mathbf{M}\|_{\mathrm{F}}^2 = n - p$. By Hanson–Wright (Theorem 3.2), conditionally on $\mathbf{X}$:

$$\mathsf{P}\big(|\mathrm{SSE} - \sigma_\varepsilon^2(n-p)| \geqslant t \;\big|\; \mathbf{X}\big) \leqslant 2\exp\left[-c\min\left(\frac{t^2}{\sigma^4(n-p)}, \frac{t}{\sigma^2}\right)\right].$$

Setting $t = \delta\sigma^2(n-p)$: $\mathsf{P}(|\mathrm{SSE} - \sigma_\varepsilon^2(n-p)| \geqslant \delta\sigma^2(n-p)) \leqslant 2\exp(-c\min(\delta^2(n-p), \delta(n-p))) \leqslant 2\exp(-c'\delta^2(n-p))$. With $\delta = C/\sqrt{n-p}$: $|\mathrm{SSE} - \sigma_\varepsilon^2(n-p)| \leqslant C\sigma^2\sqrt{n-p}$ with probability $\geqslant 1 - 2e^{-c'C^2}$. Hence $|\mathrm{MSE} - \sigma_\varepsilon^2| = |\mathrm{SSE}/(n-p) - \sigma_\varepsilon^2| \leqslant C\sigma^2/\sqrt{n-p} \to 0$.

Solution to Exercise 16.11. (a) $\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}(\mathbf{X}'\mathbf{X}\boldsymbol{\beta} + \mathbf{X}'\boldsymbol{\varepsilon})$. Adding and subtracting $\lambda\boldsymbol{\beta}$:

$$\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}[(\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})\boldsymbol{\beta} - \lambda\boldsymbol{\beta} + \mathbf{X}'\boldsymbol{\varepsilon}] = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}(\mathbf{X}'\boldsymbol{\varepsilon} - \lambda\boldsymbol{\beta}).$$

(b) $\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} - \lambda(\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}\boldsymbol{\beta}$. Squaring:

$$\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2 = \underbrace{\lambda^2\boldsymbol{\beta}'\mathbf{A}^2\boldsymbol{\beta}}_{\text{bias}^2} + \underbrace{\boldsymbol{\varepsilon}'\mathbf{X}\mathbf{A}^2\mathbf{X}'\boldsymbol{\varepsilon}}_{\text{variance}} + \text{cross term},$$

where $\mathbf{A} = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}$. The cross term vanishes in expectation since $\mathsf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$.

(c) As $p/n \to 1^-$, $\lambda_{\min}(\mathbf{X}'\mathbf{X}/n) \to 0$. OLS variance $\propto 1/\lambda_{\min}(\mathbf{X}'\mathbf{X}) \to \infty$. Ridge adds λ to every eigenvalue: $\lambda_{\min}(\mathbf{X}'\mathbf{X} + \lambda\mathbf{I}) \geqslant \lambda > 0$, controlling the variance. The cost is bias $\propto \lambda\|\boldsymbol{\beta}\|$, which is small for moderate λ.

Solution to Exercise 16.12. For Rademacher ε_i, $S_n = \sum \varepsilon_i$. The MGF is $\mathsf{E}[e^{tS_n}] = (\cosh t)^n$. By the Chernoff bound: $\mathsf{P}(S_n \geqslant s) \leqslant \inf_{t>0} e^{-ts}(\cosh t)^n$. Using $\cosh t \leqslant e^{t^2/2}$ gives $\mathsf{P}(S_n \geqslant s) \leqslant e^{-s^2/(2n)}$, matching Hoeffding. The bound $\cosh t \leqslant e^{t^2/2}$ is tight at $t = 0$ (both sides equal 1, derivatives match), so the Hoeffding bound is essentially optimal for Rademacher sums.

Solution to Exercise 16.13. (a) $\hat{\beta}_j = \mathbf{e}_j'\hat{\boldsymbol{\beta}} = \mathbf{e}_j'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = \beta_j + \mathbf{e}_j'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} = \beta_j + \mathbf{c}'\boldsymbol{\varepsilon}$ where $\mathbf{c} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{e}_j$.

(b) Conditionally on $\mathbf{X}$, $\mathbf{c}'\boldsymbol{\varepsilon} = \sum_i c_i\varepsilon_i$ is a sum of independent sub-Gaussian variables with $\|\mathbf{c}'\boldsymbol{\varepsilon}\|_{\psi_2} \leqslant C\sigma\|\mathbf{c}\|$. By the sub-Gaussian tail bound:

$$\mathsf{P}(|\hat{\beta}_j - \beta_j| \geqslant t \mid \mathbf{X}) \leqslant 2\exp\left(-\frac{ct^2}{\sigma^2\|\mathbf{c}\|^2}\right).$$

Now $\|\mathbf{c}\|^2 = \mathbf{c}'\mathbf{c} = \mathbf{e}_j'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{e}_j = [(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$.

(c) Setting $2\exp(-ct^2/(\sigma^2[(\mathbf{X}'\mathbf{X})^{-1}]_{jj})) = \alpha$ gives $t = \sigma\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj} \cdot \frac{1}{c}\ln(2/\alpha)}$. The confidence interval is:

$$\hat{\beta}_j \pm \sigma\sqrt{\frac{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}{c}\ln\frac{2}{\alpha}}.$$

Comparison. The classical interval has width $\propto t_{\alpha/2,n-p}\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$, using s in place of σ. For large $n-p$, $t_{\alpha/2,n-p} \approx z_{\alpha/2} = \Phi^{-1}(1-\alpha/2) \approx \sqrt{2\ln(2/\alpha)}$ (by the Gaussian tail approximation). So the modern and classical widths agree up to constants. The modern bound has two advantages: (1) it does not require normality, and (2) it is a finite-sample statement, not asymptotic. Its disadvantage: it uses the unknown σ rather than the data-driven s (though s concentrates around σ by Exercise 16.10).